

CURRICULUM VITAE

JAMES C. SCHNABLE

Nebraska Corn Checkoff Presidential Chair
Center for Plant Science Innovation
Department of Agronomy & Horticulture
University of Nebraska-Lincoln

Email: schnable@unl.edu
Web: schnablelab.org^a
LinkedIn: [linkedin.com/in/jamesschnable](https://www.linkedin.com/in/jamesschnable)
ORCID: [0000-0001-6739-5527](https://orcid.org/0000-0001-6739-5527)
Google Scholar: [Scholar Profile](#)

^aClickable hyperlinks are in blue throughout

Employment

University of Nebraska-Lincoln

Professor, Department of Agronomy and Horticulture	2022-Present
Nebraska Corn Checkoff Presidential Chair	2023-Present
Charles O. Gardner Professor of Agronomy	2019-2023
Associate Professor, Department of Agronomy and Horticulture	2019-2022
Assistant Professor, Department of Agronomy and Horticulture	2014-2019

X, Google, Alphabet

(Interim) Technology Lead (L7), Visiting Position During Sabbatical	2022
---	------

Danforth Plant Science Center & Chinese Academy of Agricultural Sciences

NSF PGRP Fellowship Supported Visiting Scholar	2013-2014
--	-----------

Education

University of California-Berkeley, PhD, Plant Biology	2008-2012
Cornell University, BA, Biology	2004-2008

Selected Honors and Awards

NAS Prize in Food and Agriculture Sciences National Academy of Sciences	2026
Senior Member National Academy of Inventors	2026
Outstanding Original Research Article The Plant Journal	2024
Outstanding Postdoc Mentor University of Nebraska-Lincoln	2024
Fellow Nebraska Center for Entrepreneurship	2023

Fellow PhenoRob Cluster of Excellence	2022
Paper of the Year The Plant Phenome Journal	2020
Early Career Award American Society of Plant Biologists	2019
Plant Phenotyping Early Career Award North American Plant Phenotyping Network	2019
Outstanding Scientific Article International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)	2018
Marcus Rhoades Early Career Award Maize Genetics Community	2018
Junior Faculty Excellence in Research Award University of Nebraska-Lincoln	2016

Research Support

\$30.4M in total federal funding as PI/co-PI 2015-Present
(Excludes \$40M in awards as part of two NSF Centers.)

Federal (Current)

USDA-NIFA “Sorghum grain: a domestic source of high-value natural waxes to increase harvest value and supply chain security.” (co-PI) 2025-2028 \$650k

NSF “TRTech-PGR: Digital Ideotype for Optimal Canopy Architecture” (co-PI) 2024-2027 \$2.0M

NSF “RESEARCH-PGR: Cycling to low-temperature tolerance.” (co-PI) 2024-2027 \$1.8M

USDA-NIFA “Improving Causal Gene Detection across Crop and Livestock Species.” (co-PI) 2023-2026 \$1.3M

NSF “[AI Institute for Resilient Agriculture](#)” (Investigator) 2021-2026 \$20M

Non-Federal (Current)

Nebraska Corn Board “Precision nutrition in maize: Designer arabinoxylans for modulating the human gut microbiome.” (co-PI) 2026-2027 \$376k

Nebraska Corn Board “Nebraska Nitrogen Initiative.” (PI) 2025-2027 \$450k

Nebraska Corn Board “Genomes to Fields (G2F) - Predicting Final Yield Performance in Variable Environments.” (PI) 2016-2027 \$556k (*to date*)

Completed Projects

DOE “Phenotypic and Molecular Characterization of Nitrogen Responsive Genes in Sorghum.” (co-PI) 2022-2025 \$2.7M

USDA-NIFA “[High Intensity Phenotyping Sites: Transitioning To A Nationwide Plant Phenotyping Network.](#)” (co-PI) 2020-2025 \$3M

ARPA-E “CORN (Crop Optimization Realized through Neuralnets) Demonstration” (PI) 2024-2025 \$649k

NSF “[RII Track-2 FEC: Functional analysis of nitrogen responsive networks in Sorghum.](#)” (co-PI) 2018-2024 \$4M

DOE “[TGCM: \(T\)rait, \(G\)ene, and \(C\)rop Growth \(M\)odel directed targeted gene characterization in sorghum.](#)” (PI) 2019-2024 \$2.7M

NSF “[BTT EAGER: A wearable plant sensor for real-time monitoring of sap flow and stem diameter to accelerate breeding for water use efficiency.](#)” (PI) 2019-2024 \$300k

FFAR “[Crops in silico: Increasing crop production by connecting models from the microscale to the macroscale.](#)” (co-PI) 2019-2024 \$5M

USDA-NIFA “[High Intensity Phenotyping Sites: A multi-scale, multi-modal sensing and sense-making cyber-ecosystem for Genomes to Fields.](#)” (co-PI) 2020-2024 \$2.7M

University of Nebraska “SPACE2: Space, Policy, Agriculture, Climate, and Extreme Environment.” (co-PI) 2022-2024 \$150k.

USDA-NIFA “[CPS: Medium: Field-scale, single plant-resolution agricultural management using coupled molecular and macro sensing and multi-scale data fusion and modeling](#)” (co-PI) 2020-2023 \$1.05M

ARPA-E “[Soil Organic Carbon Networked Measurement System \(SOCNET\)](#)” (co-PI) 2020-2023 \$1.9M

Wheat Innovation Foundation “A Low-Cost, High-Throughput Cold Stress Perception Assay for Sorghum Breeding.” (co-PI) 2019-2023 \$205k

ICRISAT “Identifying Novel Loci Controlling Priority Traits in Pearl Millet and Sorghum using Supervised Classification Algorithms.” (PI) 2020-2021 \$50k

ARPA-E “CORN- Crop Optimization Realized through Neuralnets” (co-PI) 2020-2022 \$620k

ARPA-E “[Low cost wireless chemical sensor networks.](#)” (co-PI) 2019-2022 \$2.2M

NSF “[Center for Root and Rhizobiome Innovation.](#)” (Investigator & Management Team Member) 2016-2021 \$20M

NSF “[RoL: FELS: EAGER: Genetic constraints on the increase of organismal complexity over time.](#)” (PI) 2018-2022 \$300k

USDA-NIFA “[Identifying mechanisms conferring low temperature tolerance in maize, sorghum, and frost tolerant relatives.](#)” (PI) 2015-2020 \$455k

ARPA-E “[In-plant and in-soil microsenors enabled high-throughput phenotyping of root nitrogen uptake and nitrogen use efficiency.](#)” (co-PI) 2017-2019 \$1.1M

USDA/NSF Joint Program “PAPM EAGER: Transitioning to the next generation plant phenotyping robots.” (co-PI) 2016-2018 \$285k

North Central Sun Grants “High throughput phenotyping to accelerate biomass sorghum improvement.” (co-PI) 2017-2019 \$193k

Daugherty Water for Food Global Institute “Optimizing the Water Use Efficiency of C4 Grain Crops Using Comparative Phenomics and Crop Models to Guide Breeding Targets.” (PI) 2017-2019 \$27k

Agricultural Research Division “A High Throughput Phenotyping Reference Dataset for GWAS in Sorghum” (PI) 2016-2018 \$100k

ICRISAT “Application of tGBS And Genomic Selection to a Hybrid Pearl Millet Breeding Program.” (PI) 2015-2017 \$45k

ConAgra “Marker Discovery & Genetic Diversity.” (replacement PI) 2014-2017 \$162k

Iowa Corn Board “Field Deployable Cameras to Quantify Dynamic Whole Plant Phenotypes in the Field.” (PI) 2014-2016 \$43k

Midwest Big Data Hub “Automatic feature extraction pipeline development for high-throughput plant phenotyping” (co-PI) 2017-2018 \$5k

Layman Award “Developing genomic tools in proso millet and comparing water use efficiency among panicoid grass crops (proso millet, corn, sorghum, foxtail millet)” (co-PI) 2014-2015 \$10k

Economic Development

Entrepreneurship

Founder, [Dryland Genetics LLC](#) 2014-Present
Uses high throughput quantitative genetics and field phenotyping technologies to develop and commercialize higher yielding cultivars of crops already naturally adapted to using little water and growing in arid regions where conventional agriculture fails in the absence of irrigation.

Co-Founder, [Data2Bio LLC](#) 2010-Present
Provides patented tGBS genotyping and genomic selection services to public and private sector plant and animal breeders in the USA and China.

Co-Founder, [EnGeniousAg LLC](#) 2017-2024
Designed, field tested, and deployed a low-cost in-field nutrient sensor to optimize nitrogen fertilizer applications. Acquired by CropX in 2024.

Entrepreneurship-Related Funding

NSF (to EnGeniousAg) “SBIR Phase II: Low-cost in-planta nitrate sensor” 2023-2024 \$1M

NSF (to EnGeniousAg) “SBIR Phase I: Low-cost in-planta nitrate sensor” 2019-2022 \$225k

USDA (to EnGeniousAg) “SBIR Phase I: Low-cost field-deployable sensors to monitor nitrate in soil and water.” 2019-2021 \$100k

Raised more than \$7M in private sector equity funding.

Industry Cooperation

Scientific Advisory Council, GeneSeek, Inc 2017-Present

Advisory Board, Afflo Sensors 2023-Present

Advisory Board, Heritable Agriculture 2025-Present

External Advisor to the Scientific Advisory Board, Indigo Agriculture 2017

External Advisor to the Scientific Advisory Board, Syngenta AG 2016

Intellectual Property

US Utility Patent 12,016,285 “Higher yielding proso (DLG240)” issued 25 June 2024 (Assignee: Dryland Genetics)

US Utility Patent 12,336,472 “Higher yielding proso (DLG40)” issued 11 June 2025 (Assignee: Dryland Genetics)

US Plant Variety Protection (PVP) Certificate 202100200 “DLG-024256 (DLG240)” issued 11 Feb 2025 (Assignee: Dryland Genetics)

US Plant Variety Protection (PVP) Certificate 202100185 “DLG-024888 (DLG40)” issued 10 Dec 2024 (Assignee: Dryland Genetics)

International Patent Application WO2025184161A1 “Higher-yielding proso varieties DLG197 and DLG317” (PCT/US2025/017324) 2025, status pending (Assignee: Dryland Genetics Inc)

US Patent Application 19/096,148 “Residuals Method to Decouple Correlated Phenotypes” 2025 (Assignee: Google)

US Patent Application 19/226,059 “Systems and Methods for Identifying DNA Sequences Regulating Pattern of Expression for Genes of Interest” 2025 (Assignee: NuTech Ventures)

US Provisional Patent Filing “A computational method for identifying specific DNA sequences that regulate the level and pattern of RNA expression for genes of interest” 2025 (Assignee: Heritable Agriculture)

US Provisional Patent Filing 63/970,172 “Portable leaf imaging chamber and image analysis methods for field phenotyping” 2026 (Assignee: NuTech Ventures)

Mentoring

Current Postdoctoral Mentees: Zhongjie Ji

Current Graduate Advisees: Waqar Ali (PhD, Complex Biosystems, US-Pakistan Knowledge Corridor Scholar), Jensina Davis (PhD, Complex Biosystems, NSF Grad Fellow), Harshita Mangal (PhD, Plant Breeding and Genetics), Karla Cuellar (PhD, Complex Biosystems), Ozgur Altundas (PhD, Agronomy), Hadiya Kounsar (MS, Agronomy)

Undergraduate Researchers: 8 NSF supported REU (Research Experience for Undergraduates) students; 3 USDA supported REEU (Research and Extension Experience for Undergraduates) students; 3 UCARE (Undergraduate Creative Activities and Research Experience) students; and 56 undergraduate students supported by regular research funding.

High School Researchers: 2 students supported through the Young Nebraska Scientist program; 1 supported by regular research funding.

Trainee Outcomes

12 former trainees are now faculty members across 4 countries: 5 in USA, 4 in China, 2 in Turkey, 1 in Poland.

Former Postdoctoral Mentees: Marcin Grzybowski (Assistant Professor, University of Warsaw), Guangchao Sun (Professor, Sichuan Agricultural University), Xiaoxi “Peggy” Meng (Assistant Professor, University of South Florida), Ranjita Thapa (Research Scientist, Inari Agriculture), Sunil KK Raju (Assistant Professor, University of California, Riverside), Ravi Mural (Assistant Professor, South Dakota State University), Jinliang Yang (Associate Professor, University of Nebraska–Lincoln), Yang Zhang (Research Scientist, St. Jude Children’s Research Hospital)

Former Visiting Scientists and Scholars: Caner Yavuz (Assistant Professor, Nigde Omer Halisdemir University, Turkey), Khang Hoang (Genetic Data Analyst, PlantEdits), Deniz Istipliler (Assistant Professor, Ege University), Lang Yan (Professor, XiChang University)

Graduated Advisees: Nikee Shrestha, PhD (Research Scientist, Bayer Crop Science), Daniel Carvalho, PhD (Postdoc, Federal University of Minas Gerais), Zhikai Liang, PhD (Assistant Professor, NDSU), Chenyong Miao, PhD (Phenomics Data Scientist, Bayer Crop Science), Nate Korth, PhD (Postdoc, North Carolina State University), Michael Tross, PhD (AI Developer, Corteva), Henrique Moura Dias, PhD (Postdoctoral Researcher, South Dakota State University), Hongyu Jin, PhD (Postdoc, South China Agricultural University), Fangyi Li, PhD (Postdoc, Kunming Institute of Botany), Xianjun Lai, PhD (Associate Professor, XiChang University), Xiuru Dai, PhD (Assistant Professor, Shandong Agricultural University), Santos Yenandy Barrera Lemus, PhD (Research Scientist, World Coffee Research), Preston Hurst, MS (Research Scientist, Syngenta), Mackenzie Zwiener, MS (Statistician, US Dept. of Homeland Security), Kyle Linders, MS (Research Agronomist, PivotBio), Ramesh Kanna Mathivanan, MS (PhD Student, Texas A&M), Sofiya Arora, MS (PhD Student, North Carolina State University), Kahheetah Barnoskie, MS, Bhushit Agarwal, MS (Software Engineer, MyVest), Srinidhi Bashyam, MS (Systems Software Developer, University of Nebraska)

Publications

H-Index: 55

Lab members in **bold**, *authors contributed equally, †undergraduate author, §corresponding author

Preprints

Grzybowski MW[§], Schnable JC (2026) Haplotype-rich cis-regulation underlies transcriptomic diversity across the breeding history of maize (*Zea mays*). *BioRxiv* doi: [10.64898/2026.02.19.706772](https://doi.org/10.64898/2026.02.19.706772)

Li H, Griffis HK, Edwards J, Flint-Garcia S, Hirsch CN, Holland JB, Knoll JE, Lima DC, de Leon N, Murray SC, Singh MP, **Schnable JC**, Sekhon RS, Thomison P, Thompson A, Wallace JG[§] (2025) Gene-by-Environment Interaction Significantly Drives Bacterial Endophyte Communities in Maize Stalks. *BioRxiv* doi: [10.64898/2025.12.23.696053](https://doi.org/10.64898/2025.12.23.696053)

Shrestha N, Ji Zhongjie, Dai X, Li P, Schnable JC[§] (2025) A sequence-based classifier distinguishes phenotype-associated genes from other gene models in plants. *BioRxiv* doi: [10.64898/2025.11.30.691407](https://doi.org/10.64898/2025.11.30.691407)

Ji Z, Ge Y, Schnable JC[§] (2025) Scalable methods for quantifying the stay green ability of corn for yield prediction by using satellite image. *AGRIxiv* doi: [10.31220/agriRxiv.2025.00339](https://doi.org/10.31220/agriRxiv.2025.00339)

Zaremejrjerdi H, Coffey L, Jubery TZ, Liu H, **Turkus J, Linders K, Schnable JC**, Schnable PS, Ganapathysubramanian B[§] (2025) MaizeEar-SAM: Zero-shot maize ear phenotyping. *ARXIV* [arXiv:2502.13399](https://arxiv.org/abs/2502.13399)

Creach M, Webster B, Newton L, **Turkus J, Schnable JC**, Thompson A, VanBuren R[§] (2025) Predicting complex phenotypes using multi-omics data in maize. *BioRxiv* doi: [10.1101/2025.09.30.679283](https://doi.org/10.1101/2025.09.30.679283)

Tross MC, Duggan G, Shrestha N, Schnable JC[§] (2024) Models trained to predict differential expression across plant organs identify distal and proximal regulatory regions. *BioRxiv* doi: [10.1101/2024.06.04.597477](https://doi.org/10.1101/2024.06.04.597477)

Faculty Publications

179. **Moura Dias H**, Sagawa GYAS, **Torres-Rodriguez JV**, **Mural RV**, **Schnable JC**, Van Sluys MA (2026) A sorghum-anchored pan-grass syntenic gene set in grasses. NAR GENOMICS AND BIOINFORMATICS doi: [10.1093/nargab/lqag030](https://doi.org/10.1093/nargab/lqag030)
178. Chen Q, Washburn JD, Lima DC ... **Schnable JC** (26th of 32 authors) ... Tuinstra MR, Wallace J, Xu W, de Leon N^S (2026) Genomes to fields 2024 maize genotype by environment prediction competition. BMC RESEARCH NOTES doi: [10.1186/s13104-026-07629-5](https://doi.org/10.1186/s13104-026-07629-5)
177. Mannon PJ^S, Benson AK, **Schnable JC**, Ding W, Juritsch AF, Rose DJ (2025) Food as Medicine: Enhancing Crop Breeding and Food Processing to Shift the Gut Microbiome for Prevention and Treatment of Chronic Diseases. THE JOURNAL OF NUTRITION doi: [10.1016/j.tjnut.2025.101298](https://doi.org/10.1016/j.tjnut.2025.101298)
176. Engelhorn J, Snodgrass S, Kok A, Seetharam A, Schneider M, Kiwit T, Singh A, Banf M, Khaiphoburch M, Runcie D, Camargo V, **Torres-Rodriguez JV**, **Sun G**, Stam M, Fiorani F, **Schnable JC**, Bass H, Hufford M, Stich B, Frommer W, Ross-Ibarra J, Hartwig T^S (2025) Phenotypic variation in maize can be largely explained by genetic variation at transcription factor binding sites. NATURE GENETICS doi: [10.1038/s41588-025-02246-7](https://doi.org/10.1038/s41588-025-02246-7) BIORxIV doi: [10.1101/2023.08.08.551183](https://doi.org/10.1101/2023.08.08.551183)
Selected as an Editor's Choice by MaizeGDB Editorial Board August 2025
175. **Mangal H**, **Linders K**, **Turkus J**, **Shrestha N**, Long B, Kuang X, Cebert E, **Torres-Rodriguez JV**, **Schnable JC**^S (2025) Genes and pathways determining flowering time variation in temperate adapted sorghum. THE PLANT JOURNAL doi: [10.1111/tjp.70250](https://doi.org/10.1111/tjp.70250)
174. **Shrestha N**^{*}, Powadi A^{*}, **Davis J**, Ayanlade TT, Liu H, **Tross MC**, **Mathivanan RK**, Bares J, **Lopez-Corona L**, **Turkus J**, Coffey L, Jubery TZ, Ge Y, Sarkar S, **Schnable JC**^S, Ganapathysubramanian B^S, Schnable PS^S (2025) Plot-level satellite imagery can substitute for UAVs in assessing maize phenotypes across multistate field trials. PLANTS, PEOPLE, PLANET doi: [10.1002/ppp3.10613](https://doi.org/10.1002/ppp3.10613) AGRIRxiv doi: [10.31220/agriRxiv.2024.00251](https://doi.org/10.31220/agriRxiv.2024.00251)
Cover Article, June 2025 doi: [10.1002/ppp3.10528](https://doi.org/10.1002/ppp3.10528)
173. Van Haute MJ, Yang Q, **Korth N**, Happ M, Kok CR, **Miao C**, Clarke J, Karnik K, Eskridge K, Urrea CA, Hyten DL, **Schnable JC**, Rose D, Benson A^S (2025) Genetic variation and historical breeding patterns in common bean (*Phaseolus vulgaris* L.) affect fermentation patterns by the human gut microbiome. COMMUNICATIONS BIOLOGY doi: [10.1038/s42003-025-09089-2](https://doi.org/10.1038/s42003-025-09089-2)
172. Powadi A, Jubery TZ, **Tross MC**, **Shrestha N**, Coffey L, **Schnable JC**, Schnable PS, Ganapathysubramanian B^S (2025) Enhancing yield prediction from plot-level satellite imagery through genotype and environment feature disentanglement. FRONTIERS IN PLANT SCIENCE doi: [10.3389/fpls.2025.1617831](https://doi.org/10.3389/fpls.2025.1617831)
171. **Jin H**, Park A, Sreedasyam A, Li G, Ge Y, Swaminathan K, Schmutz J, Clemente TE, **Schnable JC**, Yang J^S (2025) Nitrogen response and growth trajectory of sorghum CRISPR-Cas9 mutants using high-throughput phenotyping. GENOMICS COMMUNICATIONS doi: [10.48130/gcomm-0025-0011](https://doi.org/10.48130/gcomm-0025-0011)
170. **Ali W**, **Grzybowski M**, **Torres-Rodriguez JV**, **Li F**, **Shrestha N**, **Mathivanan RK**, de Bernardeaux G, Hoang K, **Mural R**, Roston RL, **Schnable JC**^S, Sahay S^S (2025) Quantitative genetics of photosynthetic trait variation in maize. JOURNAL OF EXPERIMENTAL BOTANY doi: [10.1093/jxb/eraf198](https://doi.org/10.1093/jxb/eraf198)
169. **Istipliler D**, **Tross MC**, **Bouwens B**, **Jin H**, Ge Y, Yang J, **Mural RV**, **Schnable JC**^S (2025) Heritability, heterosis, and hybrid/inbred classification ability of maize leaf hyperspectral signals under changing soil nitrogen. CROP SCIENCE doi: [10.1002/csc2.70073](https://doi.org/10.1002/csc2.70073)
"Leaf Spectra Reveal Maize Hybrid Vigor" feature in CSA News CSA News October 2025
168. **Davis JM**, Coffey LM, **Turkus J**, **López-Corona L**, **Linders K**, **Ullagaddi C**, Santra DK, Schnable PS, **Schnable JC**^S (2025) Assessing the impact of yield plasticity on hybrid performance in maize. PHYSIOLOGIA PLANTARUM doi: [10.1111/pp1.70278](https://doi.org/10.1111/pp1.70278)

167. **Davis JM**, Gaillard M, **Tross MC**, **Shrestha N**, Ostermann I, **Grove RJ**, Li B, Benes B, **Schnable JC**^S (2025) 3D reconstruction enables high-throughput phenotyping and quantitative genetic analysis of phyllotaxy. *PLANT PHENOMICS* doi: [10.1016/j.plaphe.2025.100023](https://doi.org/10.1016/j.plaphe.2025.100023)
166. Ojeda-Rivera JO, Barnes AC, Ainsworth EA ... **Schnable JC** (48th of 56 authors) ... Zambrano MA, Zhai J, Zou J, Buckler ES^S (2025) Designing a nitrogen-efficient cold-tolerant maize for modern agricultural systems. *THE PLANT CELL* doi: [10.1093/plcell/koaf139](https://doi.org/10.1093/plcell/koaf139)
165. Ozair F, Adak A, Murray SC^S, Alpers RT, Aviles AC, Lima DC, Edwards J, Ertl D, Gore MA, Hirsch CN, Knoll JE, **Schnable JC**, Singh MP, Sparks EE, Thompson A, Weldekidan T, Xu W (2025) Phenotypic plasticity in maize grain yield: Genetic and environmental insights of response to environmental gradients. *THE PLANT GENOME* doi: [10.1002/tpg2.70078](https://doi.org/10.1002/tpg2.70078)
164. Izere P, Zhao B, **Schnable JC**, Ge Y, Shi Y^S (2025) Positioning accuracy of RTK-GNSS-enabled drones and their performance in agricultural crop sensing. *JOURNAL OF THE ASABE* doi: [10.13031/ja.16306](https://doi.org/10.13031/ja.16306)
163. Quach TN, Guo M, Nazarenus T, Sato SJ, Wang M, Zhang T, Nersesian N, Ge Z, **Ullagaddi C**, Yu B, **Schnable JC**, and Clemente TE^S (2025) In context promoter bashing of the *Sorghum bicolor* gene models functionally annotated as bundle sheath cell preferred expressing phosphoenolpyruvate carboxykinase and alanine aminotransferase. *CROP SCIENCE* doi: [10.1002/csc2.70039](https://doi.org/10.1002/csc2.70039)
162. **Shrestha N**, Mangal H, **Torres-Rodriguez JV**, **Tross MC**, **Lopez-Corona L**, **Linders K**, **Sun G**, **Mural RV**, **Schnable JC**^S (2025) Off-the-shelf image analysis models outperform human visual assessment in identifying genes controlling seed color variation in sorghum. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.70013](https://doi.org/10.1002/ppj2.70013) *BIORxiv* doi: [10.1101/2024.07.22.604683](https://doi.org/10.1101/2024.07.22.604683)
161. Shrestha K, Thapa K, Kaler E, Taniguchi M, Sattler SE, **Schnable JC**, Louis J^S (2025) Employing spectral features to accelerate sorghum phenotyping against sap-feeding aphids. *PLANT DIRECT* doi: [10.1002/pld3.70092](https://doi.org/10.1002/pld3.70092)
160. **Li F**, **Grzybowski M**, Roston RL, **Schnable JC**^S (2025) Nighttime fluorescence phenotyping reduces environmental variability for photosynthetic traits and enables the identification of candidate loci in maize. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2025.1595339](https://doi.org/10.3389/fpls.2025.1595339)
159. **Mathivanan RK**, **Pedersen C**, **Turkus J**, **Shrestha N**, **Ali W**, **Torres-Rodriguez JV**, **Mural RV**, **Obata T**, **Schnable JC**^S (2025) Transcripts and genomic intervals associated with variation in metabolite abundance in maize leaves under field conditions. *BMC GENOMICS* doi: [10.1186/s12864-025-11580-3](https://doi.org/10.1186/s12864-025-11580-3)
158. **Torres-Rodriguez JV**, Li D, **Schnable JC**^S (2025) Evolving best practices for transcriptome-wide association studies accelerate discovery of gene-phenotype links. *CURRENT OPINION IN PLANT BIOLOGY* doi: [10.1016/j.pbi.2024.102670](https://doi.org/10.1016/j.pbi.2024.102670)
157. Singh A, Newton L, **Schnable JC**, Thompson AM^S (2025) Unveiling shared genetic regulators for plant architectural and biomass yield traits in sorghum. *JOURNAL OF EXPERIMENTAL BOTANY* doi: [10.1093/jxb/eraf012](https://doi.org/10.1093/jxb/eraf012) *BIORxiv* doi: [10.1101/2024.03.13.584802](https://doi.org/10.1101/2024.03.13.584802)
156. Boatwright L, Thudi M, Sangireddy MKR, Coffin AW, Tadesse HK, Vutla S, Harris-Shultz K, Knoll JE, Cuevas H, Kumar N, Soman C, **Schnable JC**, Punhuri SM^S (2024) GWAS analysis for plant height and stem diameter in sorghum using multiple phenotyping approaches. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.70008](https://doi.org/10.1002/ppj2.70008)
155. Powadi A, Jubery TZ, **Tross M**, **Schnable JC**, Ganapathysubramanian B^S (2024) Disentangling genotype and environment specific latent features for improved trait prediction using a compositional autoencoder. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2024.1476070](https://doi.org/10.3389/fpls.2024.1476070)
154. Wei J, Guo T, Mu Q, Alladassi M, **Mural RV**, Boyles RE, Hoffman L, Hayes CM, **Sigmon B**, Thompson AM, Salas-Fernandez M, Rooney WL, Kresovich S, **Schnable JC**, Li X, Yu J^S (2024) Genetic and environmental patterns underlying phenotypic plasticity in flowering time and plant height in sorghum. *PLANT, CELL AND ENVIRONMENT* doi: [10.1111/pce.15213](https://doi.org/10.1111/pce.15213)

153. Delen Y, **Mural RV**, Xu G, Delen SP, **Schnable JC**, Dweikat I, Yang J^S (2024) Dissecting the genetic architecture of sunflower disc diameter using genome-wide association study. *PLANT DIRECT* doi: [10.1002/pld3.70010](https://doi.org/10.1002/pld3.70010)
152. Ostermann I, Benes B^S, Gaillard M, Li B, **Davis J**, **Grove RJ**, **Shrestha N**, **Tross MC**, **Schnable JC** (2024) Sorghum segmentation and leaf counting using an *in silico* trained deep neural model. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.70002](https://doi.org/10.1002/ppj2.70002)
151. **Moura Dias H**, de Teledo NA, **Mural RV**, **Schnable JC**, Van Sluys MA^S (2024) TH1 gene evolutionary trends: A comprehensive plant-focused Assessment via data Mining and large-scale analysis. *GENOME BIOLOGY AND EVOLUTION* doi: [10.1093/gbe/evae212](https://doi.org/10.1093/gbe/evae212)
150. Sahay S*, **Shrestha N***, **Moura Dias H**, **Mural RV**, **Grzybowski M**, **Schnable JC**^S, Glowacka K^S (2024) Nonphotochemical quenching kinetics GWAS in sorghum identifies genes that may play conserved roles in maize and Arabidopsis thaliana photoprotection. *THE PLANT JOURNAL* doi: [10.1111/tpj.16967](https://doi.org/10.1111/tpj.16967)
149. Bai G, Barker B, Scoby D, Irmak S, Luck JD, Neale CMU, **Schnable JC**, Awada T, Ge Y^S (2024) High-throughput physiological phenotyping of crop evapotranspiration at the plot scale. *FIELD CROP RESEARCH* doi: [10.1016/j.fcr.2024.109507](https://doi.org/10.1016/j.fcr.2024.109507)
148. **Li D**, Wang Q, Tian Y, Lyu X, Zhang H, Sun Y, Hong H, Gao H, Li Y, Zhao C, Wang J, Wang R, Yang J, Liu B, Schnable PS, **Schnable JC**^S, Li Y^S, Qiu L^S (2024) TWAS facilitates gene-scale trait genetic dissection through gene expression, structural variations, and alternative splicing in soybean. *PLANT COMMUNICATIONS* doi: [10.1016/j.xplc.2024.101010](https://doi.org/10.1016/j.xplc.2024.101010)
147. **Korth N**, Yang Q, Van Haute MJ, **Tross MC**, Peng B, **Shrestha N**, **Zwiener M**, **Mural RV**, **Schnable JC**, Benson AK^S (2024) Genomic co-localization of variation affecting agronomic and human gut microbiome traits in a meta-analysis of diverse sorghum G3 doi: [10.1093/g3journal/jkae145](https://doi.org/10.1093/g3journal/jkae145) *BIORxIV* doi: [10.1101/2023.09.20.558616](https://doi.org/10.1101/2023.09.20.558616)
146. Khound R, Rajput SG, **Schnable JC**, Mani V, Dipak SK^S (2024) Genome-wide association study reveals marker-trait associations for major agronomic traits in proso millet (*Panicum miliaceum* L.). *PLANTA* doi: [10.1007/s00425-024-04465-4](https://doi.org/10.1007/s00425-024-04465-4)
145. Jia Y, **Ullagaddi C**, Allen C, Rausch K, Singh V, **Schnable JC**, Kamruzzaman M^S (2024) Optimizing feature selection with gradient boosting machines in PLS regression for predicting moisture and protein in multi-Country corn kernels via NIR spectroscopy. *FOOD CHEMISTRY* doi: [10.1016/j.foodchem.2024.140062](https://doi.org/10.1016/j.foodchem.2024.140062)
144. **Raju SKK**, **Zhang Y**, Mahboub S, Ngu DW, Qiu Y, Harmon FG, **Schnable JC**^S, Roston RL^S (2024) Rhythmic lipid and gene expression responses to chilling in panicoid grasses. *JOURNAL OF EXPERIMENTAL BOTANY* doi: [10.1093/jxb/erae247](https://doi.org/10.1093/jxb/erae247) *BIORxIV* doi: [10.1101/2023.09.29.560160](https://doi.org/10.1101/2023.09.29.560160)
143. **Tross MC**, **Grzybowski M**, Jubery TZ, **Grove RJ**[†], **Nishimwe AV**[†], **Torres-Rodriguez JV**, **Sun G**, Ganapathysubramanian B, Ge Y, **Schnable JC**^S (2024) Data driven discovery and quantification of hyperspectral leaf reflectance phenotypes across a maize diversity panel. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.20106](https://doi.org/10.1002/ppj2.20106) *BIORxIV* doi: [10.1101/2023.12.15.571950](https://doi.org/10.1101/2023.12.15.571950)
142. **Torres-Rodriguez JV**, Li D, Turkus J, Newton L, **Davis J**, **Lopez-Corona L**, **Ali W**, **Sun G**, **Mural RV**, **Grzybowski M**, Zamft B, Thompson AM, **Schnable JC**^S (2024) Population level gene expression can repeatedly link genes to functions in maize. *THE PLANT JOURNAL* doi: [10.1111/tpj.16801](https://doi.org/10.1111/tpj.16801) *BIORxIV* doi: [10.1101/2023.10.31.565032](https://doi.org/10.1101/2023.10.31.565032)
SEB-Wiley-TPJ Outstanding Original Article Award, 2024
141. **Jin H**, **Tross MC**, Tan R, Newton L, **Mural RV**, Yang J, Thompson AM, **Schnable JC**^S (2024) Imitating the “breeder’s eye”: predicting grain yield from measurements of non-yield traits. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.20102](https://doi.org/10.1002/ppj2.20102)
“Computers Learn to Predict Yield the Same Way Plant Breeders Do” feature in CSA News CSA News August 2024

140. Sahay S, **Grzybowski M**, **Schnable JC**, Glowacka K[§] (2024) Genotype-specific nonphotochemical quenching responses to nitrogen deficit are linked to chlorophyll a to b ratios. *JOURNAL OF PLANT PHYSIOLOGY* doi: [10.1016/j.jplph.2024.154261](https://doi.org/10.1016/j.jplph.2024.154261)
139. Zarei A[§], Li B, **Schnable JC**, Lyons E, Pauli D, Benes B, Barnard K (2024) PlantSegNet: 3D point cloud instance segmentation of nearby plant organs with identical semantics. *COMPUTERS AND ELECTRONICS IN AGRICULTURE* doi: [10.1016/j.compag.2024.108922](https://doi.org/10.1016/j.compag.2024.108922)
138. Rodene E, Fernando GD, Piyush V, Ge Y, **Schnable JC**, Ghosh S, Yang J[§] (2024) Image filtering to improve maize tassel detection accuracy using machine learning algorithms. *SENSORS* doi: [10.3390/s24072172](https://doi.org/10.3390/s24072172)
137. **Linders KM**, Santra D, **Schnable JC**, **Sigmon B**[§] (2024) Variation in leaf chlorophyll concentration in response to nitrogen application across maize hybrids in contrasting environments. *MICROPUBLICATION BIOLOGY* doi: [10.17912/micropub.biology.001115](https://doi.org/10.17912/micropub.biology.001115)
136. **Sun G**, Yu H, Wang P, Lopez-Guerrero MG, **Mural RV**, **Mizero ON**[†], **Grzybowski M**, Song B, van Dijk K, Schachtman DP, Zhang C, **Schnable JC**[§] (2023) A role for heritable transcriptomic variation in maize adaptation to temperate environments. *GENOME BIOLOGY* doi: [10.1186/s13059-023-02891-3](https://doi.org/10.1186/s13059-023-02891-3) *Selected as an Editor's Choice by MaizeGDB Editorial Board* August 2023
135. **Grzybowski M**[§], **Mural RV**, Xu G, **Turkus J**, Yang Jinliang, **Schnable JC** (2023) A common resequencing-based genetic marker dataset for global maize diversity. *THE PLANT JOURNAL* doi: [10.1111/tpj.16123](https://doi.org/10.1111/tpj.16123) *Cover Article, March 2023 "Research Highlight"* doi: [10.1111/tpj.16123](https://doi.org/10.1111/tpj.16123) *Selected as an Editor's Choice by MaizeGDB Editorial Board* December 2023
134. Sahay S*, **Grzybowski M***, **Schnable JC**, Glowacka K[§] (2023) Genetic control of photoprotection and photosystem II operating efficiency in plants. *NEW PHYTOLOGIST* doi: [10.1111/nph.18980](https://doi.org/10.1111/nph.18980)
133. **Sun G**, Wase N, Shu S, Jenkins J, Zhou B, Chen C, Sandor L, Plott C, Yoshinga Y, Daum C, Qi P, Barry K, Lipzen A, Berry L, Gottilla T, **Foltz A**[†], Yu H, O'Malley R, Zhang C, Devos KM, **Sigmon B**, Yu B, Obata T, Schmutz J[§], **Schnable JC**[§] (2023) Genome of *Paspalum vaginatum* and the role of trehalose mediated autophagy in increasing maize biomass. *NATURE COMMUNICATIONS* doi: [10.1038/s41467-022-35507-8](https://doi.org/10.1038/s41467-022-35507-8) *BioRxiv* doi: [10.1101/2021.08.18.456832](https://doi.org/10.1101/2021.08.18.456832) *"Research Highlight" in Nature Plants* doi: [10.1038/s41477-023-01343-x](https://doi.org/10.1038/s41477-023-01343-x)
132. DiMario R, Kophs A, Apalla A, **Schnable JC**, Cousins A[§] (2023) Multiple highly expressed phosphoenolpyruvate carboxylase genes have divergent enzyme kinetic properties in two C₄ grasses. *ANNALS OF BOTANY* doi: [10.1093/aob/mcad116](https://doi.org/10.1093/aob/mcad116)
131. Barnes AC, Myers JL, Surber SM, **Liang Z**, Mower JP, **Schnable JC**, Roston RL[§] (2023) Oligogalactolipid production during cold challenge is conserved in early diverging lineages. *JOURNAL OF EXPERIMENTAL BOTANY* doi: [10.1093/jxb/erad241](https://doi.org/10.1093/jxb/erad241)
130. Chen J, Wang Z, Tan K, Huang W, Shi J, Li T, Hu J, Wang K, Xin B, Zhao H, Song W, Hufford MB, **Schnable JC**, Ware DH, Jin W, Lai J[§] (2023) A complete telomere-to-telomere assembly of the maize genome. *NATURE GENETICS* doi: [10.1038/s41588-023-01419-6](https://doi.org/10.1038/s41588-023-01419-6) *Selected as an Editor's Choice by MaizeGDB Editorial Board* August 2023
129. Kick D, Wallace J, **Schnable JC**, Kolkman JM, Alaca B, Beissinger TM, Ertl D, Flint-Garcia S, Gage JL, Hirsch CN, Knoll JE, de Leon N, Lima DC, Moreta D, Singh MP, Weldekidan T, Washburn JD[§] (2023) Yield prediction through integration of genetic, environment, and management data through deep learning. *G3* doi: [10.1093/g3journal/jkad006](https://doi.org/10.1093/g3journal/jkad006) *BioRxiv* doi: [10.1101/2022.07.29.502051](https://doi.org/10.1101/2022.07.29.502051)
128. Chen C, Rutkoski J, **Schnable JC**, Murray S, Wang L, Jin X, Stich B, Crossa J, Hayes BJ, Zhang Z (2023) Role of the genomics–phenomics–agronomy paradigm in plant breeding. *PLANT BREEDING REVIEWS* doi: [10.1002/9781119874157.ch10](https://doi.org/10.1002/9781119874157.ch10)

127. Lima DC[§], Aviles AC, Alpers RT ... **Schnable JC** (26th of 37 authors) ... Wisser RJ, Xu W, de Leon N (2023) 2018–2019 field seasons of the Maize Genomes to Fields (G2F) G x E project. BMC GENOMIC DATA doi: [10.1186/s12863-023-01129-2](https://doi.org/10.1186/s12863-023-01129-2)
126. Wijewardane NK, Zhang H, Yang J, **Schnable JC**, Schachtman DP, Ge Y[§] (2023) A leaf-level spectral library to support high throughput plant phenotyping: Predictive accuracy and model transfer. JOURNAL OF EXPERIMENTAL BOTANY doi: [10.1093/jxb/erad129](https://doi.org/10.1093/jxb/erad129)
125. Lima DC, Aviles AC, Alpers RT ... **Schnable JC** (24th of 35 authors) ... Weldekidan T, Xu W, de Leon N[§] (2023) 2020–2021 field seasons of Maize GxE project within the Genomes to Fields Initiative. BMC RESEARCH NOTES doi: [10.1186/s13104-023-06430-y](https://doi.org/10.1186/s13104-023-06430-y)
124. Lima DC[§], Washburn JD, Varela JI ... **Schnable JC** (25th of 34 authors) ... Tuinstra MR, Wallace J, Weldekidan T, Xu W, de Leon N (2023) Genomes to fields 2022 maize genotype by environment prediction competition. BMC RESEARCH NOTES doi: [10.1186/s13104-023-06421-z](https://doi.org/10.1186/s13104-023-06421-z)
123. Gaillard M, Benes B, Tross MC, **Schnable JC** (2023) Multi-view triangulation without correspondences. COMPUTERS AND ELECTRONICS IN AGRICULTURE doi: [10.1016/j.compag.2023.107688](https://doi.org/10.1016/j.compag.2023.107688)
122. Grzybowski M[§], Zwiener M, Jin H, Wijewardane NK, Atefi A, Naldrett MJ, Alvarez S, Ge Y, **Schnable JC** (2022) Variation in morpho-physiological and metabolic responses to low nitrogen stress across the sorghum association panel. BMC PLANT BIOLOGY doi: [10.1186/s12870-022-03823-2](https://doi.org/10.1186/s12870-022-03823-2) [10.1101/2022.06.08.495271](https://doi.org/10.1101/2022.06.08.495271)
121. Yang Q, Van Haute M, Korth N, Sattler S, Toy J, Rose D, **Schnable JC**, Benson A (2022) Genetic analysis of seed traits in Sorghum bicolor that affect the human gut microbiome. NATURE COMMUNICATIONS doi: [10.1038/s41467-022-33419-1](https://doi.org/10.1038/s41467-022-33419-1)
"In Brief" in Nature Reviews Genetics doi: [10.1038/s41576-022-00543-z](https://doi.org/10.1038/s41576-022-00543-z)
"Genome Watch" in Nature Reviews Microbiology doi: [10.1038/s41579-022-00850-6](https://doi.org/10.1038/s41579-022-00850-6)
120. Li D, Bai D, Tian Y, Li Y, Zhao C, Wang Q, Gou S, Gu Y, Luan X, Wang R, Yang J, Hawkesford MJ, **Schnable JC**, Jin X, Qiu L (2022) Time series canopy phenotyping enables the identification of genetic variants controlling dynamic phenotypes in soybean. JOURNAL OF INTEGRATIVE PLANT BIOLOGY doi: [10.1111/jipb.13380](https://doi.org/10.1111/jipb.13380)
119. Khound R, Sun G, Mural RV, **Schnable JC**, Santra D[§] (2022) SNP Discovery in Proso millet (*Panicum miliaceum* L.) using low-pass genome sequencing. PLANT DIRECT doi: [10.1002/pld3.447](https://doi.org/10.1002/pld3.447)
118. Zhang K, Yang Y, Zhang X, Zhang L, Fu Y, Guo Z, Chen S, Wu J, **Schnable JC**, Yi K, Wang X, Cheng F[§] (2022) The genome of *Orychophragmus violaceus* provides genomic insights into the evolution of Brassicaceae polyploidization and its distinct traits. PLANT COMMUNICATIONS doi: [10.1016/j.xplc.2022.100431](https://doi.org/10.1016/j.xplc.2022.100431)
117. Mural RV, Sun G, Grzybowski M, Tross MC, Jin H, Smith C, Newton L, Andorf CM, Woodhouse MR, Thompson AM, Sigmon B, **Schnable JC**[§] (2022) Association mapping across a multitude of traits collected in diverse environments identifies pleiotropic loci in maize. GIGASCIENCE doi: [10.1093/gigascience/giac080](https://doi.org/10.1093/gigascience/giac080) [10.1101/2022.02.25.480753](https://doi.org/10.1101/2022.02.25.480753)
116. Meier M, Xu G, Lopez-Guerrero, Li G, Smith C, Sigmon B, Herr J, Alfano J, Ge Y, **Schnable JC**, Yang J[§] (2022) Maize root-associated microbes likely under adaptive selection by the host to enhance phenotypic performance. eLIFE doi: [10.7554/eLife.75790](https://doi.org/10.7554/eLife.75790)
115. Korth N, Parsons L, Van Haute M, Yang Q, Hurst JP, **Schnable JC**, Holding DR, Benson AK[§] (2022) The unique seed protein composition of quality protein popcorn promotes growth of beneficial bacteria from the human gut microbiome. FRONTIERS IN MICROBIOLOGY doi: [10.3389/fmicb.2022.921456](https://doi.org/10.3389/fmicb.2022.921456)
114. Mural RV, **Schnable JC**[§] (2022) Can the grains offer each other helping hands? Convergent molecular mechanisms associated with domestication and crop improvement in rice and maize. MOLECULAR PLANT doi: [10.1016/j.molp.2022.04.003](https://doi.org/10.1016/j.molp.2022.04.003)
Peer Reviewed Invited Perspective

113. Boatwright JL, Sapkota S, **Jin H, Schnable JC**, Brenton Z, Boyles R, Kresovich S[§] (2022) Sorghum Association Panel whole-genome sequencing establishes cornerstone resource for dissecting genomic diversity. *THE PLANT JOURNAL* doi: [10.1111/tpj.15853](https://doi.org/10.1111/tpj.15853) *BIORxIV* doi: [10.1101/2021.12.22.473950](https://doi.org/10.1101/2021.12.22.473950)
112. Rodene E, Xu G, Delen SP, **Smith C**, Ge Y, **Schnable JC**, Yang J[§] (2022) A UAV-based high-throughput phenotyping approach to assess time-series nitrogen responses and identify traits associated genetic components in maize. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.20030](https://doi.org/10.1002/ppj2.20030) *BIORxIV* doi: [10.1101/2021.05.24.445447](https://doi.org/10.1101/2021.05.24.445447)
111. Yu H, Sandhu J, **Sun G**, Nguyen H, Clemente T, **Schnable JC**, Walia H, Xie W, Yu B, Mower JP, Zhang C[§] (2022) Pervasive misannotation of the smallest microexons that are evolutionarily conserved and crucial for gene function in plants. *NATURE COMMUNICATIONS* doi: [10.1038/s41467-022-28449-8](https://doi.org/10.1038/s41467-022-28449-8)
110. **Tross MC[‡]**, Gaillard M, **Zwiener M[‡]**, **Miao C**, **Grove RJ**, Li B, Benes B, **Schnable JC[§]** (2021) 3D reconstruction identifies loci linked to variation in angle of individual sorghum leaves. *PEERJ* doi: [10.7717/peerj.12628](https://doi.org/10.7717/peerj.12628) *BIORxIV* doi: [10.1101/2021.06.15.448566](https://doi.org/10.1101/2021.06.15.448566)
109. Diao X[§], Zhang H, Tang S, **Schnable JC**, He Q, Gao Y, Luo M, Jia G, Feng B, Zhi H (2021) Genome-Wide DNA polymorphism analysis and molecular marker development of *Setaria italica* variety 'SSR41' and application in positional cloning of *Setaria* white leaf sheath gene SiWLS1. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2021.743782](https://doi.org/10.3389/fpls.2021.743782)
108. **Miao C**, **Guo A[‡]**, Thompson AM, Yang J, Ge Y, **Schnable JC[§]** (2021) Automation of leaf counting in maize and sorghum using deep learning. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.20022](https://doi.org/10.1002/ppj2.20022) *BIORxIV* doi: [10.1101/2020.12.19.423626](https://doi.org/10.1101/2020.12.19.423626)
107. **Sun G[§]**, **Mural RV**, **Turkus JD**, **Schnable JC** (2021) Quantitative resistance loci to southern rust mapped in a temperate maize diversity panel. *PHYTOPATHOLOGY* doi: [10.1094/PHYTO-04-21-0160-R](https://doi.org/10.1094/PHYTO-04-21-0160-R) *BIORxIV* doi: [10.1101/2021.04.02.438220](https://doi.org/10.1101/2021.04.02.438220)
106. **Mural RV**, **Grzybowski M**, **Miao C**, **Damke A[‡]**, Sapkota S, Boyles RE, Salas Fernandez MG, Schnable PS, **Sigmon B**, Kresovich S, **Schnable JC[§]** (2021) Meta-analysis identifies pleiotropic loci controlling phenotypic trade-offs in sorghum. *GENETICS* doi: [10.1093/genetics/iyab087](https://doi.org/10.1093/genetics/iyab087) *BIORxIV* doi: [10.1101/2020.10.27.355495](https://doi.org/10.1101/2020.10.27.355495)
Cover Article, July 2021 (Vol. 218, Issue 3) <https://doi.org/10.1093/genetics/iyab087>
105. **Grzybowski M**, Wijewardane NK, Atefi A, Ge Y, **Schnable JC[§]** (2021) The potential of hyperspectral reflectance as a tool for quantitative genetics in crops. *PLANT COMMUNICATIONS* doi: [10.1016/j.xplc.2021.100209](https://doi.org/10.1016/j.xplc.2021.100209)
104. Zhou Y, Kusmec A, Mirnezami SV, Srinivasan L, Jubery TZ, **Schnable JC**, Salas-Fernandez MG, Nettleton D, Ganapathysubramanian B, Schnable PS[§] (2021) Identification and utilization of genetic determinants of trait measurement errors in image-based, high-throughput phenotyping. *THE PLANT CELL* doi: [10.1093/plcell/koab134](https://doi.org/10.1093/plcell/koab134)
103. Atefi A, Ge Y[§], Pitla S, **Schnable JC** (2021) Robotic Technologies for High-Throughput Plant Phenotyping: Reviews and Perspectives. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2021.611940](https://doi.org/10.3389/fpls.2021.611940)
102. Alzadjali A, Veeranampalayam-Sivakumar A, Alali MH, Deogun JS, Scott S, **Schnable JC**, Shi Y[§] (2021) Maize tassel detection from UAV imagery using deep learning. *FRONTIERS IN ROBOTICS AND AI* [10.3389/frobt.2021.600410](https://doi.org/10.3389/frobt.2021.600410)
101. Meier MA, Lopenz-Guerrero MG, Guo M, Schmer MR, Herr JR, **Schnable JC**, Alfano JR, Yang J[§] (2021) Rhizosphere microbiomes in a historical maize/soybean rotation system respond to host species and nitrogen fertilization at genus and sub-genus levels. *APPLIED AND ENVIRONMENTAL MICROBIOLOGY* doi: [10.1128/AEM.03132-20](https://doi.org/10.1128/AEM.03132-20) *BIORxIV* doi: [10.1101/2020.08.10.244384](https://doi.org/10.1101/2020.08.10.244384)
100. Serb DD, **Meng X**, **Schnable JC**, Bashir E, Michaud JP, Vara Prasad PV, Perumal R (2021) Comparative transcriptome analysis reveals genetic mechanisms of sugarcane aphid resistance in grain sorghum. *INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES* doi: [10.3390/ijms22137129](https://doi.org/10.3390/ijms22137129)

99. Hurst JP, **Schnable JC**, Holding DR^S (2021) Tandem duplicate expression patterns are conserved between maize haplotypes of the α -zein gene family. *PLANT DIRECT* doi: [10.1002/pld3.346](https://doi.org/10.1002/pld3.346)
98. Busta L, Schmitz E, Kosma D, **Schnable JC**, Cahoon EB^S (2021) A co-opted steroid synthesis gene, maintained in sorghum but not maize, is associated with a divergence in leaf wax chemistry. *PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA* doi: [10.1073/pnas.2022982118](https://doi.org/10.1073/pnas.2022982118)
97. **Meng X, Liang Z, Dai X, Zhang Y**, Mahboub S, Ngu DW[‡], Roston RL, **Schnable JC**^S (2021) Predicting transcriptional responses to cold stress across plant species. *PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA*. doi: [10.1073/pnas.2026330118](https://doi.org/10.1073/pnas.2026330118) *BIORxiv* doi: [10.1101/2020.08.25.266635](https://doi.org/10.1101/2020.08.25.266635)
96. Sankaran S^S, Marzougui A, **Hurst JP**, Zhang C, **Schnable JC**, Shi Y (2021) Can high resolution satellite imagery be used in high-throughput field phenotyping? *TRANSACTIONS OF THE ASABE* doi: [10.13031/trans.14197](https://doi.org/10.13031/trans.14197)
95. Zhu Y, Chen Y, Ali Md. A, Dong L, Wang X, Archontoulis SV, **Schnable JC**, Castellano MJ^S (2021) Continuous in situ soil nitrate sensors: a comparison with conventional measurements and the value of high temporal resolution measurements. *SOIL SCIENCE SOCIETY OF AMERICA JOURNAL* doi: [10.1002/saj2.20226](https://doi.org/10.1002/saj2.20226)
Recipient of the Soil Science Society of America Best Paper Award in 2023
94. **Lai X**, Bendix C, **Zhang Y**, **Schnable JC**, Harmon FG^S (2021) 72-hour diurnal RNA-seq analysis of fully expanded third leaves from maize, sorghum, and foxtail millet at 3-hour resolution. *BMC RESEARCH NOTES* doi: [10.1186/s13104-020-05431-5](https://doi.org/10.1186/s13104-020-05431-5)
93. Rogers AR, Dunne JC, Romay C ... **Schnable JC** (24th of 39 authors) ... Kaeppler S, De Leon N, Holland JB^S (2021) The importance of dominance and genotype-by-environment interactions on grain yield variation in a large-scale public cooperative maize experiment. *G3: GENES | GENOMES | GENETICS* doi: [10.1093/g3journal/jkaa050](https://doi.org/10.1093/g3journal/jkaa050)
Selected as an Editor's Choice by MaizeGDB Editorial Board February 2021
92. Jarquin D, de Leon N, Romay C ... **Schnable JC** (24th of 33 authors) ... Wissner RJ, Xu W, Lorenz A (2021) Utility of climatic information via combining ability models to improve genomic prediction for yield within the Genomes to Fields maize project. *FRONTIERS IN GENETICS* doi: [10.3389/fgene.2020.592769](https://doi.org/10.3389/fgene.2020.592769)
91. DiMario RJ, Kophs AN, Pathare VS, **Schnable JC**, Cousins AB^S (2021) Phosphoenolpyruvate carboxylase kinetic variation provides opportunity to enhance C4 photosynthetic efficiency. *THE PLANT JOURNAL* doi: [10.1111/tpj.15141](https://doi.org/10.1111/tpj.15141)
90. Thudi M, Palakurthi R, **Schnable JC**, Chitikineni A, Dreisigacker S, Mace E, Srivastava RK, Satyavathi CT, Odeny D, Tiwari VK, Lam HM, Hong YB, Singh VK, Li G, Xu Y, Chen X, Nguyen H, Sivasankar S, Close TJ, Stein N, Jackson SA, Shubo W, Varshney RK^S (2021) Genomic resources in plant breeding for sustainable agriculture. *JOURNAL OF PLANT PHYSIOLOGY* doi: [10.1016/j.jplph.2020.153351](https://doi.org/10.1016/j.jplph.2020.153351)
89. **Raju SKK**, Atkins M[‡], **Enerson A**[‡], **Carvalho DS**, Studer AJ, Ganapathysubramanian B, **Schnable PS**, **Schnable JC**^S (2020) Leaf Angle eXtractor - A high throughput image processing framework for leaf angle measurement in maize and sorghum. *APPLICATIONS IN PLANT SCIENCES* doi: [10.1002/aps3.11385](https://doi.org/10.1002/aps3.11385)
88. Gaillard M*, **Miao C***, **Schnable JC**^S, Benes B^S (2020) Voxel carving based 3D reconstruction of sorghum identifies genetic determinants of radiation interception efficiency. *PLANT DIRECT* doi: [10.1002/pld3.255](https://doi.org/10.1002/pld3.255) *BIORxiv* doi: [10.1101/2020.04.06.028605v1](https://doi.org/10.1101/2020.04.06.028605v1)
87. **Raju SKK**^S, Thompson AM, **Schnable JC** (2020) Advances in plant phenomics: From data and algorithms to biological insights. *APPLICATIONS IN PLANT SCIENCES* doi: [10.1002/aps3.11386](https://doi.org/10.1002/aps3.11386)
86. Atefi A, Ge Y^S, Pitla S, **Schnable JC** (2020) Robotic detection and grasp of maize and sorghum: stem measurement with contact. *ROBOTICS* doi: [10.3390/robotics9030058](https://doi.org/10.3390/robotics9030058)

85. Wang R, Qiu Y^S, Zhou Y, **Liang Z, Schnable JC** (2020) A high-throughput phenotyping pipeline for image processing and functional growth curve analysis. *PLANT PHENOMICS* doi: [10.34133/2020/7481687](https://doi.org/10.34133/2020/7481687)
84. **Lai X**, Bendix C, **Yan L, Zhang Y, Schnable JC**, Harmon F^S (2020) Interspecific analysis of diurnal gene regulation in panicoid grasses identifies known and novel regulatory motifs. *BMC GENOMICS* doi: [10.1186/s12864-020-06824-3](https://doi.org/10.1186/s12864-020-06824-3)
Selected as an Editor's Choice by MaizeGDB Editorial Board October 2020
83. Han J, Wang P, Wang Q, Lin Q, Yu G, **Miao C**, Dao Y, Wu R, **Schnable JC**, Tang H, Wang K^S (2020) Genome-wide characterization of DNase I-hypersensitive sites and cold response regulatory landscapes in grasses. *THE PLANT CELL* doi: [10.1105/tpc.19.00716](https://doi.org/10.1105/tpc.19.00716)
"In Brief" highlighting this article by SKK Raju doi: [10.1105/tpc.20.00471](https://doi.org/10.1105/tpc.20.00471)
82. Moiseyev G, Park K, Cui X, Freitas D, Rajagopa D, Konda A, Martin-Olenski M, Mcham M, Liu K, Du Q, **Schnable JC**, Moriyama E, Cahoon E, Chi Z^S (2020) RGPDB: Database of root-associated genes and promoters in maize, soybean, and sorghum. *DATABASE* doi: [10.1093/database/baaa038](https://doi.org/10.1093/database/baaa038)
81. **Miao C**, Xu Y, Liu S, Schnable PS, **Schnable JC**^S (2020) Increased power and accuracy of causal locus identification in time-series genome-wide association in sorghum. *PLANT PHYSIOLOGY* doi: [10.1104/pp.20.00277](https://doi.org/10.1104/pp.20.00277) *BIORxIV* doi: [10.1101/2020.02.16.951467](https://doi.org/10.1101/2020.02.16.951467)
"News and Views" highlighting this article by Y Yu doi: [10.1104/pp.20.00797](https://doi.org/10.1104/pp.20.00797)
80. **Dai X**, Xu Z, **Liang Z**, Tu X, Zhong S, **Schnable JC**^S, Li P^S (2020) Non-homology based prediction of gene functions. *THE PLANT GENOME* doi: [10.1002/tpg2.20015](https://doi.org/10.1002/tpg2.20015) *BIORxIV* doi: [10.1101/730473](https://doi.org/10.1101/730473)
79. Peng B, Guan K^S, Ainsworth EA, Asseng S, Bernacchi CJ, Cooper M, Delucia EH, Elliot JW, Ewert F, Grant RF, Gustafson DI, Hammer GL, Jin Z, Jones JW, Kimm H, Lawrence DM, Li Y, Lombardozzi DL, Marshall-Colon A, Messina CD, Ort DR, **Schnable JC**, Tang J, Vallejos CE, Wu A, Yin X, Zhou W (2020) Advancing multi-scale crop modeling for agricultural climate change adaptation assessment. *NATURE PLANTS* doi: [10.1038/s41477-020-0625-3](https://doi.org/10.1038/s41477-020-0625-3)
78. Adams J, Qiu Y^S, Xu Y, **Schnable JC** (2020) Plant segmentation by supervised machine learning methods. *THE PLANT PHENOME JOURNAL* doi: [10.1002/ppj2.20001](https://doi.org/10.1002/ppj2.20001)
77. **Liang Z**, Qiu Y, **Schnable JC**^S (2020) Distinct characteristics of genes associated with phenome-wide variation in maize (*Zea mays*). *MOLECULAR PLANT* doi: [10.1016/j.molp.2020.03.003](https://doi.org/10.1016/j.molp.2020.03.003) *BIORxIV* doi: [10.1101/534503](https://doi.org/10.1101/534503)
Selected as an Editor's Choice by MaizeGDB Editorial Board May 2020
76. Benes B, Guan K, Lang M, Long S, Lynch J, Marshall-Colon A^S, Peng B, **Schnable JC**, Sweetlove L, Turk M (2020) Multiscale computational models can guide experimentation and targeted measurements for crop improvement. *THE PLANT JOURNAL* doi: [10.1111/tpj.14722](https://doi.org/10.1111/tpj.14722)
75. McFarland BA, AlKhalifah N, Bohn ... **Schnable JC** (34 of 54 authors) ... Xu W, Yeh CT, de Leon N^S (2020) Maize Genomes to Fields (G2F): 2014 –2017 field seasons' genotype, phenotype, climatic, soil and inbred ear image datasets. *BMC RESEARCH NOTES* doi: [10.1186/s13104-020-4922-8](https://doi.org/10.1186/s13104-020-4922-8)
74. **Carvalho DS**, Nishimwe AV[†], **Schnable JC**^S (2020) IsoSeq transcriptome assembly of C3 panicoid grasses provides tools to study evolutionary change in the Panicoideae. *PLANT DIRECT* [10.1002/pld3.203](https://doi.org/10.1002/pld3.203) *BIORxIV* doi: [10.1101/689356](https://doi.org/10.1101/689356)
73. **Miao C**, Pages A[†], Xu Z, Rodene E, Yang J, **Schnable JC**^S (2020) Semantic segmentation of sorghum using hyperspectral data identifies genetic associations. *PLANT PHENOMICS* doi: [10.34133/2020/4216373](https://doi.org/10.34133/2020/4216373)
72. Jarquin D^S, Howard R, **Liang Z**, Gupta SK, **Schnable JC**, Crossa J (2020) Enhancing hybrid prediction in pearl millet using genomic and/or multi-environment phenotypic information of inbreds. *FRONTIERS IN GENETICS* doi: [10.3389/fgene.2019.01294](https://doi.org/10.3389/fgene.2019.01294)

71. Zheng Z, Hey S, Jubery T, Liu T, Yang Y, Coffey L, **Miao C, Sigmon B, Schnable JC**, Hochholdinger F, Ganapathysubramanian B, Schnable PS[§] (2020) Shared genetic control of root system architecture between *Zea mays* and *Sorghum bicolor*. *PLANT PHYSIOLOGY* doi: [10.1104/pp.19.00752](https://doi.org/10.1104/pp.19.00752)
70. Qi P, Eudy D, **Schnable JC**, Schmutz J, Raymer P, Devos KM[§] (2019) High density genetic maps of seashore paspalum using genotyping-by-sequencing and their relationship to the *Sorghum bicolor* genome. *SCIENTIFIC REPORTS* doi: [10.1038/s41598-019-48257-3](https://doi.org/10.1038/s41598-019-48257-3)
69. **Schnable JC**[§] (2019) Genes and gene models, an important distinction. *NEW PHYTOLOGIST* doi: [10.1111/nph.16011](https://doi.org/10.1111/nph.16011)
Commissioned Material: Tansley Insight Medal
68. Ge Y[§], Atefi A, Zhang H, **Miao C**, Ramamurthy RK, **Sigmon B**, Yang J, **Schnable JC** (2019) High-throughput analysis of leaf physiological and chemical traits with VIS-NIR-SWIR spectroscopy: A case study with a maize diversity panel. *PLANT METHODS* doi: [10.1186/s13007-019-0450-8](https://doi.org/10.1186/s13007-019-0450-8)
67. Ali MA, Wang X, Chen Y, Jiao Y, Mahal NK, Satyanarayana M, Castellano MJ, **Schnable JC**, Schnable PS, Dong L[§] (2019) Continuous Monitoring of Nitrate Variation Using Miniature Soil Sensor with Poly(3-octyl-thiophene) and Molybdenum Disulfide Nanocomposite. *ACS APPLIED MATERIALS & INTERFACES* doi: [10.1021/acsami.9b07120](https://doi.org/10.1021/acsami.9b07120)
66. Li Y, **Li D**, Jiao Y, **Schnable JC**, Li Y, Li H, Chen H, Hong H, Zhang T, Liu B, Liu Z, You Q, Tian Y, Gou Y, Guan R, Zhang L, Chang R, Zhang Z, Reif J, Zhou X, Schnable PS, Qiu L.[§] (2019) Identification of Loci Controlling Adaptation in Chinese Soybean Landraces via a Combination of Conventional and Bioclimatic GWAS. *PLANT BIOTECHNOLOGY JOURNAL* doi: [10.1111/pbi.13206](https://doi.org/10.1111/pbi.13206)
65. Atefi A, Ge Y[§], Pitla S, **Schnable JC** (2019) *In vivo* human-like robotic phenotyping of leaf traits in maize and sorghum. *COMPUTERS AND ELECTRONICS IN AGRICULTURE* doi: [10.1016/j.compag.2019.104854](https://doi.org/10.1016/j.compag.2019.104854)
64. Li L, Li X, Li L, **Schnable JC**, Gu R, J Wang[§] (2019) QTL identification and epistatic effect analysis of seed size- and weight-related traits in *Zea mays* L. *MOLECULAR BREEDING* doi: [10.1007/s11032-019-0981-8](https://doi.org/10.1007/s11032-019-0981-8)
63. **Yan L, Kumar SKK, Lai X, Zhang Y, Dai X**, Rodriguez O, Mahboub S, Roston RL, **Schnable JC**[§] (2019) Parallels between artificial selection in temperate maize and natural selection in the cold-adapted crop-wild relative *Tripsacum*. *THE PLANT JOURNAL* doi: [10.1111/tpj.14376](https://doi.org/10.1111/tpj.14376) *BIORxiv* doi: [10.1101/187575](https://doi.org/10.1101/187575)
62. Bai G, Ge Y[§], Scoby D, Leavit B, Irmak S, Graef G, **Schnable JC**, Awada T. (2019) NU-Spidercam: A large-scale, cable-driven, integrated sensing and robotic system for precision phenotyping, remote sensing, and agronomic research. *COMPUTERS AND ELECTRONICS IN AGRICULTURE* doi: [10.1016/j.compag.2019.03.009](https://doi.org/10.1016/j.compag.2019.03.009)
61. Zou C, Miki D, **Li D**, Tang Q, Xiao L, **Rajput S**, Deng P, Peng L, Huang R, Zhang M, Sun Y, Hu J, Fu X, Schnable P, Li F, Zhang H, Feng B, Zhu X, Liu R, **Schnable JC**, Zhu JK, Zhang H[§] (2019) The genome of broomcorn millet. *NATURE COMMUNICATIONS* doi: [10.1038/s41467-019-08409-5](https://doi.org/10.1038/s41467-019-08409-5)
60. **Miao C, Yang J, Schnable JC**[§] (2018) Optimizing the identification of causal variants across varying genetic architectures in crops. *PLANT BIOTECHNOLOGY JOURNAL* doi: [10.1111/pbi.13023](https://doi.org/10.1111/pbi.13023) *BIORxiv* doi: [10.1101/310391](https://doi.org/10.1101/310391)
59. Ott A, **Schnable JC**, Yeh CT, Wu L, Liu C, Hu HC, Dolgard CL, Sarkar S, Schnable PS[§] (2018) Linked read technology for assembling large complex and polyploid genomes. *BMC GENOMICS* doi: [10.1186/s12864-018-5040-z](https://doi.org/10.1186/s12864-018-5040-z)
58. Liu S,* **Schnable JC**,* Ott A,* Yeh CT, Springer NM, Yu J, Meuhbauer G, Timmermans MCP, Scanlon MJ, Schnable PS[§] (2018) Intragenic Meiotic Crossovers Generate Novel Alleles with Transgressive Expression Levels. *MOLECULAR BIOLOGY AND EVOLUTION* doi: [10.1093/molbev/msy174](https://doi.org/10.1093/molbev/msy174)

57. **Raju SKK**, Barnes A, **Schnable JC**, Roston RL[§] (2018) Low-temperature tolerance in land plants: Are transcript and membrane responses conserved? *PLANT SCIENCE* doi: [10.1016/j.plantsci.2018.08.002](https://doi.org/10.1016/j.plantsci.2018.08.002)
56. **Miao C**, Fang J, Li D, Liang P, Zhang X, **Yang J**, **Schnable JC**, Tang H[§] (2018) Genotype-Corrector: improved genotype calls for genetic mapping. *SCIENTIFIC REPORTS* doi: [10.1038/s41598-018-28294-0](https://doi.org/10.1038/s41598-018-28294-0)
55. Alkhalifah N, Campbell DA, Falcon CM, ... **Schnable JC** (31 of 44 authors) ... Spalding EP, Edwards J, Lawrence-Dill CJ[§] (2018) Maize Genomes to Fields: 2014 and 2015 field season genotype, phenotype, environment, and inbred ear image datasets. *BMC RESEARCH NOTES* doi: [10.1186/s13104-018-3508-1](https://doi.org/10.1186/s13104-018-3508-1)
54. **Liang Z**, Gupta SK, Yeh CT, **Zhang Y**, **Ng DW**[†], Kumar R, Patil HT, Mungra KD, Yadav DV, Rathore A, Srivastava RK, Gupta R, **Yang J**, Varshney RK, Schnable PS, **Schnable JC**[§] (2018) Phenotypic data from inbred parents can improve genomic prediction in pearl millet hybrids. *G3: GENES GENOMES GENETICS* doi: [10.1534/g3.118.200242](https://doi.org/10.1534/g3.118.200242)
Selected as the outstanding scientific article of 2018 by ICRISAT's research program in Asia.
53. Xu Y, Qiu Y, **Schnable JC**[§] (2018) Functional modeling of plant growth dynamics. *THE PLANT PHENOME JOURNAL* doi: [10.2135/tppj2017.09.0007](https://doi.org/10.2135/tppj2017.09.0007) *BIORxIV* doi: [10.1101/190967](https://doi.org/10.1101/190967)
Received the "Outstanding Paper Award" from TPPJ editorial board in 2020.
52. **Carvalho DS**, **Schnable JC**, Almeida AMR[§] (2018) Integrating phylogenetic and network approaches to study gene family evolution: the case of the AGAMOUS family of floral genes. *EVOLUTIONARY BIOINFORMATICS* doi: [10.1177/1176934318764683](https://doi.org/10.1177/1176934318764683) *BIORxIV* doi: [10.1101/195669](https://doi.org/10.1101/195669)
51. **Lai X**, **Yan L**, Lu Y, **Schnable JC**[§] (2018) Largely unlinked gene sets targeted by selection for domestication syndrome phenotypes in maize and sorghum. *THE PLANT JOURNAL* doi: [10.1111/tpj.13806](https://doi.org/10.1111/tpj.13806) *BIORxIV* doi: [10.1101/184424](https://doi.org/10.1101/184424)
50. **Liang Z**, **Schnable JC**[§] (2017) Functional divergence between subgenomes and gene pairs after whole genome duplications. *MOLECULAR PLANT* doi: [10.1016/j.molp.2017.12.010](https://doi.org/10.1016/j.molp.2017.12.010)
49. **Liang Z**, Pandey P, Stoerger V, Xu Y, Qiu Y, Ge Y, **Schnable JC**[§] (2017) Conventional and hyperspectral time-series imaging of maize lines widely used in field trials. *GIGA SCIENCE* doi: [10.1093/giga-science/gix117](https://doi.org/10.1093/giga-science/gix117) *BIORxIV* doi: [10.1101/169045](https://doi.org/10.1101/169045)
48. Gage J, Jarquin D, Romay M, ... **Schnable JC** (29th of 40 authors) ... Yu J, de Leon N[§] (2017) The effect of artificial selection on phenotypic plasticity in maize. *NATURE COMMUNICATIONS* doi: [10.1038/s41467-017-01450-2](https://doi.org/10.1038/s41467-017-01450-2)
Selected as an Editor's Choice by MaizeGDB Editorial Board December 2017
47. Washburn JD, **Schnable JC**, Brutnell TP, Shao Y, **Zhang Y**, Ludwig M, Davidse G, Pires JC[§] (2017) Genome-guided phylo-transcriptomic methods and the nuclear phylogenetic tree of the Paniceae grasses. *SCIENTIFIC REPORTS* doi: [10.1038/s41598-017-13236-z](https://doi.org/10.1038/s41598-017-13236-z)
46. Ott A,* Liu S,* **Schnable JC**, Yeh CT, Wang C, Schnable PS[§] (2017) Tunable Genotyping-By-Sequencing (tGBS®) enables reliable genotyping of heterozygous loci. *NUCLEIC ACIDS RESEARCH* doi: [10.1093/nar/gkx853](https://doi.org/10.1093/nar/gkx853)
45. **Lai X**, **Schnable JC**, Liao Z, Xu J, Zhang G, Li C, Hu E, Rong T, Xu Y, Lu Y[§] (2017) Genome-wide characterization of non-reference transposable elements insertion polymorphisms reveals genetic diversity in tropical and temperate maize. *BMC GENOMICS* doi: [10.1186/s12864-017-4103-x](https://doi.org/10.1186/s12864-017-4103-x)
44. Mei W, Boatwright L, Feng G, **Schnable JC**, Barbazuk WB[§] (2017) Evolutionarily conserved alternative splicing across monocots. *GENETICS* doi: [10.1534/genetics.117.300189](https://doi.org/10.1534/genetics.117.300189)
Cover Article October 2017 Issue
43. Pandey P, Ge Y[§], Stoerger V, **Schnable JC** (2017) High throughput in vivo analysis of plant leaf chemical properties using hyperspectral imaging. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2017.01348](https://doi.org/10.3389/fpls.2017.01348)

42. **Zhang Y, Ngu DW[†], Carvalho D, Liang Z, Qiu Y, Roston RL, Schnable JC[§]** (2017) Differentially regulated orthologs in sorghum and the subgenomes of maize. *THE PLANT CELL* doi: [10.1105/tpc.17.00354](https://doi.org/10.1105/tpc.17.00354) *Selected as an Editor's Choice by MaizeGDB Editorial Board* August 2017
41. **Lai X,* Behera S,* Liang Z, Lu Y, Deogun JS, Schnable JC[§]** (2017) STAG-CNS: An order-aware conserved noncoding sequence discovery tool for arbitrary numbers of species. *MOLECULAR PLANT*. doi: [10.1016/j.molp.2017.05.010](https://doi.org/10.1016/j.molp.2017.05.010)
40. **Lai X, Schnable JC[§]** (2017) Harnessing the potential of the tea tree genome. *MOLECULAR PLANT*. doi: [10.1016/j.molp.2017.05.009](https://doi.org/10.1016/j.molp.2017.05.009)
39. Mei W, Liu S, **Schnable JC**, Yeh C, Springer NM, Schnable PS, Barbazuk WB[§] (2017) A comprehensive analysis of alternative splicing in paleopolyploid maize. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2017.00694](https://doi.org/10.3389/fpls.2017.00694)
38. Joyce BL, Huag-Baltzell A, Davey S, Bomhoff M, **Schnable JC**, Lyons E[§] (2016) FractBias: a graphical tool for assessing fractionation bias after whole genome duplications. *BIOINFORMATICS* doi: [10.1093/bioinformatics/btw666](https://doi.org/10.1093/bioinformatics/btw666)
37. Walley JW,* Sartor RC,* Shen Z, Schmitz RJ, Wu KJ, Urich MA, Nery JR, Smith LG, **Schnable JC**, Ecker JR, Briggs SP[§] (2016) Integration of omic networks in a developmental atlas of maize. *SCIENCE* doi: [10.1126/science.aag1125](https://doi.org/10.1126/science.aag1125) *Selected as an Editor's Choice by MaizeGDB Editorial Board* September 2016
36. Ge Y[§], Bai G, Stoerger V, **Schnable JC** (2016) Temporal dynamics of maize plant growth, water use, and plant water content using automated high throughput RGB and hyperspectral imaging. *COMPUTERS AND ELECTRONICS IN AGRICULTURE* doi: [10.1016/j.compag.2016.07.028](https://doi.org/10.1016/j.compag.2016.07.028)
35. **Liang Z, Schnable JC[§]** (2016) RNA-seq based analysis of population structure within the maize inbred B73. *PLOS ONE* doi: [10.1371/journal.pone.0157942](https://doi.org/10.1371/journal.pone.0157942)
34. Rajput SG, Santra DK[§], **Schnable JC** (2016) Mapping QTLs for morpho-agronomic traits in proso millet (*Panicum miliaceum* L.). *MOLECULAR BREEDING* doi: [10.1007/s11032-016-0460-4](https://doi.org/10.1007/s11032-016-0460-4)
33. Chao S, Wu J, Liang J, **Schnable JC**, Yang W, Cheng F, Wang X[§] (2016) Impacts of whole genome triplication on MIRNA evolution in *Brassica rapa*. *GENOME BIOLOGY AND EVOLUTION* doi: [10.1093/gbe/evv206](https://doi.org/10.1093/gbe/evv206)
32. Tang H, Bomhoff MD, Briones E, **Schnable JC**, Lyons E[§] (2015) SynFind: compiling syntenic regions across any set of genomes on demand. *GENOME BIOLOGY AND EVOLUTION* doi: [10.1093/gbe/evv219](https://doi.org/10.1093/gbe/evv219)
31. Washburn JD, **Schnable JC**, Davidse G, Pires JC[§] (2015) Phylogeny and photosynthesis of the grass tribe Paniceae. *AMERICAN JOURNAL OF BOTANY* doi: [10.3732/ajb.1500222](https://doi.org/10.3732/ajb.1500222)
30. Tang H, Zhang X, **Miao C**, Zhang J, Ming R, **Schnable JC**, Schnable PS, Lyons E, Lu J[§] (2015) ALLMAPS: robust scaffold ordering based on multiple maps. *GENOME BIOLOGY* doi: [10.1186/s13059-014-0573-1](https://doi.org/10.1186/s13059-014-0573-1)
29. **Schnable JC[§]** (2015) Genome evolution in maize: from genomes back to genes. *ANNUAL REVIEW OF PLANT BIOLOGY* doi: [10.1146/annurev-arplant-043014-115604](https://doi.org/10.1146/annurev-arplant-043014-115604)
28. Paschold A, Larson NB, Marcon C, **Schnable JC**, Yeh C, Lanz C, Nettleton D, Piepho H, Schnable PS, Hochholdinger F[§] (2014) Nonsyntenic genes drive highly dynamic complementation of gene expression in maize hybrids. *PLANT CELL* doi: [10.1105/tpc.114.130948](https://doi.org/10.1105/tpc.114.130948)

Postdoctoral Publications

27. Weissmann S, Huang P, Wiechert M, Furoyama K, Brutnell TP, Taniguchi M, **Schnable JC**,[§] Mockler TC[§] (2021) DCT4 - a new member of the dicarboxylate transporter family in C₄ grasses. *GENOME BIOLOGY AND EVOLUTION* doi: [10.1093/gbe/evaa251](https://doi.org/10.1093/gbe/evaa251) *BIORxIV* doi: [10.1101/762724](https://doi.org/10.1101/762724)
26. Nani TF, **Schnable JC**, Washburn JD, Albert P, Pereira WA, Sobrinho FS, Birchler JA, Techia VH[§] (2018). Location of low copy genes in chromosomes of *Brachiaria* spp. *MOLECULAR BIOLOGY REPORTS* doi: [10.1007/s11033-018-4144-5](https://doi.org/10.1007/s11033-018-4144-5)
25. Studer AJ*, **Schnable JC***, Weissmann S, Kolbe AR, McKain MR, Shao Y, Cousins AB, Kellogg EA, Brutnell TP[§] (2016) The draft genome of *Dichanthelium oligosanthes*: A C3 panicoid grass species. *GENOME BIOLOGY* doi: [10.1186/s13059-016-1080-3](https://doi.org/10.1186/s13059-016-1080-3)
24. Huang P, Studer AJ, **Schnable JC**, Kellogg EA, Brutnell TP[§] (2016) Cross species selection scans identify components of C4 photosynthesis in the grasses. *JOURNAL OF EXPERIMENTAL BOTANY* doi: [10.1093/jxb/erw256](https://doi.org/10.1093/jxb/erw256)
"Insight" highlighting this article by PA Christin also published in JXB doi: [10.1093/jxb/erw390](https://doi.org/10.1093/jxb/erw390)
23. Liu X, Tang S, Jia G, **Schnable JC**, Su X, Tang C, Zhi H, Diao X[§] (2016) The C-terminal motif of SiAGO1b is required for the regulation of growth, development and stress responses in foxtail millet [*Setaria italica* (L.) P. Beauv.]. *JOURNAL OF EXPERIMENTAL BOTANY* doi: [10.1093/jxb/erw135](https://doi.org/10.1093/jxb/erw135)
22. Jia G, Liu X, **Schnable JC**, Niu Z, Wang C, Li Y, Wang Sh, Wang Su, Liu J, Gou E, Diao X[§] (2015) Microsatellite variations of elite *Setaria* varieties released during last six decades in China. *PLOS ONE* doi: [10.1371/journal.pone.0125688](https://doi.org/10.1371/journal.pone.0125688)
21. Qie L, Jia G, Zhang W, **Schnable JC**, Shang Z, Li W, Liu B, Li M, Chai, Y, Zhi H, Diao X[§] (2014) Mapping of quantitative trait loci (QTLs) that contribute to germination and early seedling drought tolerance in the interspecific cross *Setaria italica* x *Setaria viridis*. *PLOS ONE* doi: [10.1371/journal.pone.0101868](https://doi.org/10.1371/journal.pone.0101868)
20. Diao X[§], **Schnable JC**, Bennetzen JL, Li J[§] (2014) Initiation of *Setaria* as a model plant. *FRONTIERS OF AGRICULTURAL SCIENCE AND ENGINEERING* doi: [10.15302/J-FASE-2014011](https://doi.org/10.15302/J-FASE-2014011)

Graduate Publications

19. Woodhouse MR[§], Sen S, Schott D, Portwood JL, Walley JL, Andorf CM, **Schnable JC** (2021) qTeller: A tool for comparative multi-genomic gene expression analysis. *BIOINFORMATICS* doi: [10.1093/bioinformatics/btab604](https://doi.org/10.1093/bioinformatics/btab604)
18. Cheng F, Sun C, Wu J, **Schnable JC**, Woodhouse MR, Liang J, Cai C, Freeling M,[§] Wang X[§] (2016) Epigenetic regulation of subgenome dominance following whole genome triplication in *Brassica rapa*. *NEW PHYTOLOGIST* doi: [10.1111/nph.13884](https://doi.org/10.1111/nph.13884)
17. Almeida AMR, Yockteng R, **Schnable JC**, Alvarez-Buylla ER, Freeling M, Specht CD[§] (2014) Co-option of the polarity gene network shapes filament morphology in angiosperms. *SCIENTIFIC REPORTS* doi: [10.1038/srep06194](https://doi.org/10.1038/srep06194)
16. Martin JA, Johnson NV, Gross SM, **Schnable JC**, Meng X, Wang M, Coleman-Derr D, Lindquist E, Wei C, Kaeppler S, Chen F, Wang Z[§] (2014) A near complete snapshot of the *Zea mays* seedling transcriptome revealed from ultra-deep sequencing. *SCIENTIFIC REPORTS* doi: [10.1038/srep04519](https://doi.org/10.1038/srep04519)
Selected as an Editor's Choice by MaizeGDB Editorial Board May 2014
15. Garsmeur O,* **Schnable JC***, Almeida A, Jourda C, D'Hont A,[§] Freeling M[§] (2014) Two evolutionarily distinct classes of paleopolyploidy. *MOLECULAR BIOLOGY AND EVOLUTION* doi: [10.1093/molbev/mst230](https://doi.org/10.1093/molbev/mst230)

14. Turco G, **Schnable JC**, Pedersen B, Freeling M^S (2013) Automated conserved noncoding sequence (CNS) discovery reveals differences in gene content and promoter evolution among the grasses. *FRONTIERS IN PLANT SCIENCES* doi: [10.3389/fpls.2013.00170](https://doi.org/10.3389/fpls.2013.00170)
13. **Schnable JC**, Wang X, Pires JC, Freeling M^S (2012) Escape from preferential retention following repeated whole genome duplication in plants. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2012.00094](https://doi.org/10.3389/fpls.2012.00094)
12. Freeling M^S, Woodhouse MR, Subramaniam S, Turco G, Lisch D, **Schnable JC** (2012) Fractionation mutagenesis and similar consequences of mechanisms removing dispensable or less-expressed DNA in plants. *CURRENT OPINION IN PLANT BIOLOGY* doi: [10.1016/j.pbi.2012.01.015](https://doi.org/10.1016/j.pbi.2012.01.015)
11. Tang H^S, Woodhouse MR, Cheng F, **Schnable JC**, Pedersen BS, Conant GC, Wang X, Freeling M, Pires JC (2012) Altered patterns of fractionation and exon deletions in *Brassica rapa* support a two-step model of paleohexaploidy. *GENETICS* doi: [10.1534/genetics.111.137349](https://doi.org/10.1534/genetics.111.137349)
10. **Schnable JC**, Freeling M, Lyons E^S (2012) Genome-wide analysis of syntenic gene deletion in the grasses. *GENOME BIOLOGY AND EVOLUTION* doi: [10.1093/gbe/evs009](https://doi.org/10.1093/gbe/evs009)
Selected as an Editor's Choice by MaizeGDB Editorial Board Dec 2012
9. Zhang W, Wu Y, **Schnable JC**, Zeng Z, Freeling M, Crawford GE, Jiang J^S (2012) High-resolution mapping of open chromatin in the rice genome. *GENOME RESEARCH* doi: [10.1101/gr.131342.111](https://doi.org/10.1101/gr.131342.111)
8. Eichten SR,* Swanson-Wagner RA,* **Schnable JC**, Waters AJ, Hermanson PJ, Liu S, Yeh C, Jia Y, Gendler K, Freeling M, Schnable PS, Vaughn MW, Springer NM^S (2011) Heritable epigenetic variation among maize inbreds. *PLoS GENETICS* doi: [10.1371/journal.pgen.1002372](https://doi.org/10.1371/journal.pgen.1002372)
Selected as an Editor's Choice by MaizeGDB Editorial Board Jan 2012
Recommended by Faculty of 1000
7. **Schnable JC**, Lyons E^S (2011) Comparative genomics with maize and other grasses: from genes to genomes. *MAYDICA* 56(1763) 77-93 [Link directly to PDF](#)
6. Tang H, Lyons E, Pedersen B, **Schnable JC**, Paterson AH, Freeling M. (2011) Screening synteny blocks in pairwise genome comparisons through integer programming. *BMC BIOINFORMATICS* doi: [10.1186/1471-2105-12-102](https://doi.org/10.1186/1471-2105-12-102)
5. **Schnable JC**, Pedersen BS, Subramaniam S, Freeling M^S (2011) Dose-sensitivity, conserved noncoding sequences and duplicate gene retention through multiple tetraploidies in the grasses. *FRONTIERS IN PLANT SCIENCE* doi: [10.3389/fpls.2011.00002](https://doi.org/10.3389/fpls.2011.00002)
Commentary by Birchler and Veitia also published in Frontiers in Plant Science doi: [10.3389/fpls.2011.00064](https://doi.org/10.3389/fpls.2011.00064)
4. **Schnable JC**^S, Freeling M (2011) Genes identified by visible mutant phenotypes show increased bias towards one of two maize subgenomes. *PLoS ONE* doi: [10.1371/journal.pone.0017855](https://doi.org/10.1371/journal.pone.0017855)
3. **Schnable JC**, Springer NM, Freeling M^S (2011) Differentiation of the maize subgenomes by genome dominance and both ancient and ongoing gene loss. *PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES* doi: [10.1073/pnas.1101368108](https://doi.org/10.1073/pnas.1101368108)
Selected as an Editor's Choice by MaizeGDB Editorial Board May 2011
2. Woodhouse MR,* **Schnable JC**,* Pedersen BS, Lyons E, Lisch D, Subramaniam S, Freeling M^S (2010) Following tetraploidy in maize, a short deletion mechanism removed genes preferentially from one of the two homeologs. *PLoS BIOLOGY* doi: [10.1371/journal.pbio.1000409](https://doi.org/10.1371/journal.pbio.1000409)
Selected as an Editor's Choice by MaizeGDB Editorial Board August 2010
PLoS Biology Cover Article Recommended by Faculty of 1000
1. The International Brachypodium Initiative (2010) Genome sequencing and analysis of the model grass *Brachypodium distachyon*. *NATURE* doi: [10.1038/nature08747](https://doi.org/10.1038/nature08747)

Peer Reviewed Conference Papers

13. Wang X, Hatasaka B, Liu Z, Tope S, Karkhanis M, Noh S, Sium F, **Mural RV**, Kim H, Mastrangelo C, Zang L, **Schnable JC**, Ji M (2024) SPARC-LoRa: A scalable, power-efficient, affordable, reliable, and cloud service-enabled LoRa networking system for agriculture applications. IEEE INTERNATIONAL SYMPOSIUM ON DYNAMIC SPECTRUM ACCESS NETWORKS (DySPAN) doi: [10.1109/DySPAN60163.2024.10632833](https://doi.org/10.1109/DySPAN60163.2024.10632833)
12. Noh M, Sium F, Tope S, Khan S, Karkhanis M, Wang L, Deshpande A, Dalapati R, **Mural RV**, Mastrangelo C, Zang L, Ji M, **Schnable JC**, Kim H (2023) Localization of crop damage utilizing a wake up gas sensor network. TRANSDUCERS 2023 Kyoto, Japan
11. Khan SH, Karkhanis M, Hatasaka B, Tope S, Noh S, Bulbul A, Banerjee A, Ji M, Mastrangelo CH, Kim H, Dalapati R, Zang L, **Mural RV**, **Schnable JC**, Kim K (2022) Field deployment of a nanogap gas sensor for crop damage detection. 35TH INTERNATIONAL CONFERENCE ON MICRO ELECTRO MECHANICAL SYSTEMS CONFERENCE (MEMS) Berlin, Germany doi: [10.1109/MEMS51670.2022.9699614](https://doi.org/10.1109/MEMS51670.2022.9699614)
10. Khan SH, Tope S, Dalpati R, Kim KH, Noh M, Bulbul A, **Mural RV**, Banerjee A, **Schnable JC**, Ji M, Mastrango C, Zang L, Kim H (2021) Development of a gas sensor for green leaf volatile detection. TRANSDUCERS 2021 doi: [10.1109/Transducers50396.2021.9495597](https://doi.org/10.1109/Transducers50396.2021.9495597)
9. Gaillard M, **Miao C**, **Schnable JC**, Benes B (2020) Sorghum Segmentation by Skeleton Extraction. COMPUTER VISION PROBLEMS IN PLANT PHENOTYPING (CVPPP 2020) Glasgow, UK
8. Sankaran S, Zhang C, **Hurst JP**, Marzougui A, Sivakumar ANV, Li J, **Schnable JC**, Shi Y (2020) Investigating the potential of satellite imagery for high-throughput field phenotyping applications. SPIE DEFENSE + COMMERCIAL SENSING Online Only, California, USA doi: [10.1117/12.2558729](https://doi.org/10.1117/12.2558729)
7. Al-Zadjali A, Shi Y, Scott S, Deogun JS, and **Schnable JC** (2020) Faster-R-CNN based deep learning for locating corn tassels in UAV imagery. SPIE DEFENSE + COMMERCIAL SENSING Online Only, California, USA doi: [10.1117/12.2560596](https://doi.org/10.1117/12.2560596)
6. **Miao C**, **Pages A[†]**, Xu Z, **Schnable JC** (2019) Sorghum organ classification in hyperspectral images using supervised machine learning classification methods. SECOND INTERNATIONAL WORKSHOP ON MACHINE LEARNING FOR CYBER-AGRICULTURAL SYSTEMS (MLCAS 2019) Ames, IA, USA
5. **Askey B[†]**, Yang Q, Benson AK, **Schnable JC** (2019) Computer vision phenotyping of 371 Sorghum bicolor BTx623 x ISC3620C recombinant inbred lines for QTL detection. SECOND INTERNATIONAL WORKSHOP ON MACHINE LEARNING FOR CYBER-AGRICULTURAL SYSTEMS (MLCAS 2019) Ames, IA, USA
4. Jiao Y, Wang X, Chen Y, Castellano MJ, **Schnable JC**, Schnable PS, Dong L (2019) In-planta nitrate detection using insertable plant microsensor. 20TH INTERNATIONAL CONFERENCE ON SOLID-STATE SENSORS, ACTUATORS AND MICROSYSTEMS Berlin, Germany doi: [10.1109/TRANSDUCERS.2019.8808527](https://doi.org/10.1109/TRANSDUCERS.2019.8808527)
3. Ali MA, Wang X, Chen Y, Jiao Y, Castellano MJ, **Schnable JC**, Schnable PS, Dong L (2019) Novel all-solid-state soil nutrient sensor using nanocomposite of poly(3-octyl-thiophene) and molybdenum sulfate. 20TH INTERNATIONAL CONFERENCE ON SOLID-STATE SENSORS, ACTUATORS AND MICROSYSTEMS Berlin, Germany doi: [10.1109/TRANSDUCERS.2019.8808341](https://doi.org/10.1109/TRANSDUCERS.2019.8808341)
2. Behera S, Deogun JS, **Lai X**, **Schnable JC** (2017) B529 DiCE: Discovery of Conserved Noncoding Sequences Efficiently. IEEE BIBM 2017 Kansas City, MO, USA doi: [10.1109/BIBM.2017.8217628](https://doi.org/10.1109/BIBM.2017.8217628)
1. Chaudhury SD, Steorger V, Samal A, **Schnable JC**, **Liang Z**, Yu J (2016) [Automated vegetative stage phenotyping analysis of maize plants using visible light images](#). KDD: DATA SCIENCE FOR FOOD, ENERGY AND WATER San Francisco, CA, USA

Selected Other Publications

5. **Liang Z, Meng X, Schnable JC** (2023) A transferable machine learning framework for predicting transcriptional responses of genes across species. *Plant Gene Regulatory Networks: Methods and Protocols*. Editors: Kerstin Kaufmann and Klaas Vandepoele Publisher: Springer, New York, NY.
4. Clark J, Qiu Y, **Schnable JC**. (2019) Experimental design for controlled environment high throughput plant phenotyping. *High Throughput Plant Phenotyping: Methods and Protocols*. Editor: Argelia Lorence Publisher: Springer, New York, NY.
3. Tang H, Lyons E, **Schnable JC** (2013) Early history of the angiosperms. *Genomes of Herbaceous Land Plants*. Editor: Andrew Paterson Publisher: Academic Press, Amsterdam
2. Goff SA, **Schnable JC**, Feldmann KA (2013) The evolution of plant gene and genome sequencing. *Genomes of Herbaceous Land Plants* Editor: Andrew Paterson Publisher: Academic Press, Amsterdam
1. **Schnable JC** and Freeling M (2012) Maize (*Zea mays*) as a model for studying the impact of gene and regulatory sequence loss following whole genome duplication. *Polyploidy and Genome Evolution*. Editors: Soltis PS & Soltis DE Publisher: Springer New York, NY

Selected Recent Service

University

Department of Agronomy and Horticulture Promotion and Tenure Committee	2023-Present
Nebraska Food for Health Center Faculty Advisory Committee	2017-2026
UNL Faculty Greenhouse Committee	2015-2021
Department of Agronomy and Horticulture Awards Committee	2019-2021
Department of Agronomy and Horticulture Graduate Admissions Committee	2019-2022
Biotech Seminar Series Committee	2017-2019
Agronomy and Horticulture Faculty Advisory Committee	2017-2019
Agronomy and Horticulture Strategic Planning Committee	2018-2019
Department of Agronomy and Horticulture Peer Evaluation Committee	2016-2018
Search Committee, Nebraska EPSCoR/IDeA Director	2018
Organizing Committee “International Millet Symposium 2018”	2018
Organizing Committee “Predictive Crop Design, Genome to Phenome”	2017
Search Committee, Director of Phenomic Sciences	2017
Search Committee, Agricultural Research Division	2016
Search Committee, Quantitative Life Sciences Initiative	2016
Search Committee, Department of Agronomy and Horticulture	2016
Organizing Committee “Plant Phenomics: from pixels to traits”	2015

Professional

Associate Editor: Molecular Plant	2014-Present
Guest Editor: The Plant Cell	2019-Present
Data Management Subcommittee, Maize Genetics Research Collaboration Network	2018-2020
MaizeGDB Advisory Committee	2018-2020
Host Committee for International Plant Phenotyping Network 2024 Conference	2023-2024
Grant Reviewer: NSF (panel & ad hoc), USDA (panel), JGI (panel), Genome British Columbia (ad hoc).	
Peer Reviewer (selected, recent): Bioinformatics, BMC Genomics, BMC Plant Biology, G3: Genes Genomes Genetics, Genome Biology, Genome Biology & Evolution, Heredity, Journal of Experimental Botany, JoVE, Molecular Biology and Evolution, Molecular Plant, Nature Communications, Nature Plants, New Phytologist, Nucleic Acids Research, PeerJ, Photosynthesis Research, Physiologia Plantarum, Plant Cell, Plant Cell & Environment, Plant Direct, The Plant Genome, The Plant Journal, Plant Methods, Plant Physiology, PLoS Genetics, Proceedings of the National Academy of Sciences, Science, Science Advances	

Public Engagement

Video

Big Ten Network “B1G Impact Research” Feature on corn genetics research (2024 football season)	2024
Farm Journal: Mapping the Corn Genome	2023
RFD-TV Interview on telomere-to-telomere maize genome assembly	2023
Psomagen: A Look into Plant DNA Analysis at the Schnable Lab in Nebraska	2021
Market Journal: Sorghum as Biofuel	2019
10/11 News: Researchers aim to use FitBit technology to test water supply in corn fields	2019

Audio and Podcast

Crop Science Podcast: Corn Breeding Advances	2026
GVENCK Interview at 9th International Meeting on Plant Breeding	2025
Brownfield Ag News: University of Nebraska researcher helping develop virtual replica corn field	2024
Genome Insider Podcast: Crops as Tough as World Cup Turf	2023
Brownfield Ag News: New sensors can measure nitrates in water and growing plants	2023
KRVN Radio Interview on corn genetics	2023
1010 KSIR Farm Radio: Proso millet for the high plains	2022
Brownfield Ag News: Researchers develop Fitbit-like sensor to measure corn water use	2019
Peggy Smedley Show: IoT measures and manages plants	2018

Print and Online Stories

Hastings Tribune: Feedback from producers crucial in research decisions	2026
Lincoln Journal Star: UNL's Schnable honored by National Academy of Sciences for plant science research	2026
The Hindu: Clustering maize plants together can improve their insect resistance	2025
Lincoln Journal Star: Nebraska researchers help further understanding of how genetics affect corn development	2025
Ethanol Producer Magazine: Corn's Next Step Change	2024
Midwest Messenger: In his DNA: New Nebraska Corn Checkoff presidential chairman is leader in groundbreaking corn genetic fieldwork	2023
Nebraska Public Media: Millets — ancient drought-resistant grains — could help the Midwest survive climate change	2023
Illinois Farmer Today: Scientist develops new crop genetics for food and profit	2023
High Plains Journal: Genetic mapping offers new promise for proso millet	2019
Omaha World Herald: Nebraska, Iowa researchers developing "Fitbit for corn" to make water use more efficient	2019
Lincoln Journal Star: Robots in the cornfield? UNL team believes it's the next big thing	2019
Plantae: Faculty Job: Myths & Realities — an interview with James Schnable	2019

Invited Talks

At External Institutions

University of California-Berkeley, Berkeley, CA, USA	2025
Texas Tech, Lubbock, TX, USA	2025
The Traits Company, Durham, NC, USA	2025
North Carolina State University, Raleigh, NC, USA	2025
Bayer, St. Louis, MO USA	2025
South Dakota State University, Brookings, SD, USA	2024
University of Toronto, Toronto, Canada	2024
University of Georgia-Athens, Athens, GA, USA	2023
University of Arizona, Tucson, AZ, USA	2023
Oregon State University, Corvallis, OR, USA	2023
Carnegie Institution for Science, Stanford, CA, USA	2022
Center for Sorghum Improvement, Manhattan, KS, USA	2022 (<i>Remote</i>)
CIRAD, Montpellier, France	2022
California State East Bay, Hayward, CA, USA	2021 (<i>Remote, COVID</i>)

University of Missouri, Columbia, MO, USA	2020 (<i>Remote, COVID</i>)
Rutgers University, New Brunswick, NJ, USA	2020 (<i>Remote, COVID</i>)
Bayer Crop Science, St. Louis, MO, USA	2020 (<i>Remote, COVID</i>)
University of Bonn, Bonn, Germany	2020 (<i>Remote, COVID</i>)
King Abdullah University of Science and Technology, Jeddah, Saudi Arabia	2020 (<i>Remote, COVID</i>)
University of Hawaii, Manoa, HI, USA (<i>Brewbaker Lecture</i>)	2019
Miami University, Oxford, OH, USA	2019
University of Massachusetts Amherst, Amherst, MA, USA	2019
Cornell University, Ithaca, NY, USA	2019
Research Triangle Park, NC, USA	2018
Washington State University, Pullman, WA, USA	2018
University of Delaware, Newark, DE, USA	2018
Chinese Academy of Agricultural Sciences, Beijing, China	2017
Beijing Academy of Agricultural and Forestry Sciences, Beijing, China	2017
University of Minnesota, St. Paul, MN, USA	2017
Iowa State University, Ames, IA, USA	2017
University of Missouri-Columbia, Columbia, MO, USA	2017
Kansas State University, Manhattan, KS, USA	2016
University of Georgia-Athens, Athens, GA, USA	2016
University of California-San Diego, San Diego, CA, USA	2016
Chinese Academy of Agricultural Sciences, Beijing, China	2015
Beijing Academy of Agricultural and Forestry Sciences, Beijing, China	2015
Sichuan Agricultural University, Chengdu, China	2015
Huazhong Agricultural University, Wuhan, China	2015
Shandong Agricultural University, Tai'an, China	2015
Monsanto, St. Louis, MO, USA	2015
Henan Agricultural University, Zhengzhou, China	2014
Chinese Academy of Tropical Agriculture, Haikou, China	2014
Cornell University, Ithaca, NY, USA	2014
Interdisciplinary Plant Group Seminar Series, University of Missouri, Columbia, MO, USA	2012
Donald Danforth Plant Science Center, St. Louis, MO, USA	2012
China Agricultural University, Beijing, China	2012
Chinese Academy of Agricultural Sciences, Beijing, China	2012
MaizeGDB, Ames, IA, USA	2012
University of Arizona, Tucson, AZ, USA	2011

At External Conferences

Invited presentations only. Excludes presentations selected based on abstracts or applications.

Texas A&M Plant Science Symposium (Student Organized), College Station, TX, USA	2026
NAPPN 2026, East Lansing, MI, USA	2026
Pan-Regulatory Genomics, Plant and Animal Genome, San Diego, CA, USA	2026
Molecular Plant Science, Genomics, and AI for Research and Breeding, Plant and Animal Genome, San Diego, CA, USA	2026
Plant Cistromics, Plant and Animal Genome, San Diego, CA, USA	2026
Polish Society of Experimental Plant Biology, Warsaw, Poland	2025
KWS Vegetable Breeding and Research, Wageningen, Netherlands	2025 (Remote)
9th International Meeting on Plant Breeding (Student Organized), Piracicaba, Sao Paulo, Brazil	2025
Corteva New Frontiers Conference, Johnston, IA, USA	2025
Baker Plant Breeding Symposium (Student Organized), Ames, IA, USA	2025
Scaling Trait Discovery Session, Plant and Animal Genome, San Diego, CA, USA	2025
ARPA-E Agricultural Portfolio Summit, San Antonio, TX, USA	2024
CSU Crop Adaptation Symposium (Student Organized), Fort Collins, CO, USA	2024
West African Plant Science Symposium (Student Organized)	2024 (Remote)
CROPS, Huntsville, AL, USA	2024
Freeling Symposium, Berkeley, CA, USA	2024
Purdue Plant Science Symposium (Student Organized), West Lafayette, IN, USA	2023
IROS (Intelligent Robots and Systems), Detroit, MI, USA	2023
Sorghum in the 21st Century, Montpellier, France	2023
Corteva Symposium Series, North of Rio de Janeiro State University (Student Organized), Campos dos Goytacazes, Brazil	2023 (Remote)
Iowa Biotech Showcase, Ankeny, IA, USA	2023
SFBV (French Society of Plant Biology), Montpellier, France	2022
Plant Response to Stresses and Environmental Signals, Beijing, China	2022 (Remote)
IPPN-CEPPG Workshop on Environment Simulation and Phenotyping, Gatersleben, Germany	2022 (Remote)
Michigan State Genome Sciences Symposium (Student Organized), East Lansing, MI, USA	2022
Nexus Informatics, Kansas City, MO, USA	2022
Molecular Plant Virtual Seminar Series	2022 (Remote, COVID)
DIGICROP 2022	2022 (Remote, COVID)
Machine Learning for Cyber-Agricultural Systems (Keynote)	2021 (Remote, COVID)

Plant Science Symposium West Africa (Student Organized)	2021 (<i>Remote, COVID</i>)
Soybean Breeders Workshop	2021 (<i>Remote, COVID</i>)
NAPPN 2021	2021 (<i>Remote, COVID</i>)
DIGICROP 2020	2020 (<i>Remote, COVID</i>)
National Association of Plant Breeders Annual Meeting, Lincoln, NE, USA	2020 (<i>Remote, COVID</i>)
iGenomX Session, Plant and Animal Genome, San Diego, CA, USA	2020
Systems Biology and Ontologies Session, Plant and Animal Genome, San Diego, CA, USA	2020
Guelph Plant Sciences Symposium (Student Organized), Guelph, Ontario, Canada	2019
Future of Machine Learning for Cyber-Agricultural Systems Panel, Ames, IA, USA	2019
Gene Mapping Session, Plant and Animal Genome, San Diego, CA, USA	2019
Plant Energy Biology Forum, Perth, Australia	2018
The Plant Phenome Journal Webinar Series	2018
Entrepreneurship Panel, USDA FACT: Genomes to Fields, Ames, IA, USA	2018
Plant Phenotype Session, Plant and Animal Genome, San Diego, CA, USA	2018
Plant Genome Evolution, Sitges, Spain	2017
Purdue Plant Science Symposium (Student Organized), West Lafayette, IN, USA	2017
P ² IRC Annual Symposium, Saskatoon, Saskatchewan, Canada	2017
Maize Tools and Resources (Maize Genetics Conference pre-meeting), St. Louis, MO, USA	2017
Phenome, Tucson, AZ, USA	2017
Corn Breeding Research Meeting, Jacksonville, FL, USA	2016
Molecular Plant Symposium: From Model Species to Crops, Shanghai, China	2015
Corn Breeding Research Meeting, St. Charles, IL, USA	2015
Life Technologies Session, Plant and Animal Genome, San Diego, CA, USA	2015
Maize Session, Plant and Animal Genome, San Diego, CA, USA	2015
Millet as Crop: Past and Future, Aohan, Inner Mongolia, China	2014
Plant Genomes in China Meeting, Tai'an, China	2012
American Society of Plant Biology, Austin, TX, USA	2012
Polyploidy Session, Plant and Animal Genome, San Diego, CA, USA	2012
CSSA Translational Genomics Session, Plant and Animal Genome, San Diego, CA, USA	2012

Internal

Biological Systems Engineering Departmental Seminar Series, Lincoln, NE	2026
Nebraska Plant Science Symposium, Lincoln, NE	2026
American Society for Plant Biology Midwest, Lincoln, NE	2025
Nebraska Sorghum Symposium, Grand Island, NE	2025
Great Plains Network Annual Meeting, Lincoln, NE	2024
Nebraska Agribusiness Club, Lincoln, NE	2024
Animal Science Departmental Seminar Series, UNL	2023
CROPS Entrepreneurship/Industry Career Panel (Student Organized), UNL	2023
Complex Biosystems Seminar Series, UNL	2021 (<i>In Person, COVID</i>)
Agronomy & Horticulture Departmental Seminar Series, UNL	2020 (<i>Remote, COVID</i>)
Nebraska Plant Science Symposium (Student Organized), Lincoln, NE	2019
UNL Plant Phenomics Symposium	2018
NeDA 2017: 2nd Nebraska Data Analytics Workshop, UNL	2017
Water for Food Global Conference, UNL	2017
Complex Biosystems Seminar Series, UNL	2017
Food Science Departmental Seminar Series, UNL	2016
Animal Science Departmental Seminar Series, UNL	2016
Agronomy & Horticulture Departmental Seminar Series, UNL	2015
Plant Science Retreat, UNL	2014

Professional Memberships

American Association for the Advancement of Science
 American Society of Plant Biology
 Crop Science Society of America
 National Association of Plant Breeders
 North American Plant Phenotyping Network